



POLITECNICO MILANO 1863

School of Civil, Environmental and Land
Management Engineering

ADVANCED STRUCTURAL DESIGN
Academic year 2022/2023

TECHNICAL REPORT

OBJECT: *Structural design of Reinforced Concrete slab*

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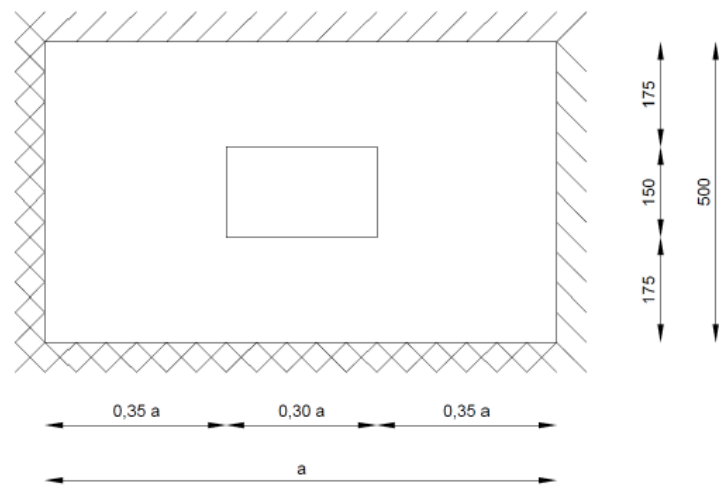
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MAIN REPORT

Section 1

Given Information

1	Requirement	The following reinforced concrete slab located in a shopping center has to be designed at the ultimate limit state under uniformly distributed loading using reinforcing steel bars class B450 and appropriate concrete class based on Euro code 2 indications.																																																				
2	Allowed Design State	Ultimate Limit State																																																				
3	Reference Code	Eurocode 2																																																				
4	Architecture of the slab and Cartesian coordinate reference system	 Figure 1 – Top view of concrete slab, dimensions in cm																																																				
5	Slab Loading (Uniformly Distributed)	• Non-Structural members: 1.0 KN/m² • Self-weight: To be computed in the following section • Live load: To be computed in the following section based on this formula ($q_k = 5.0 * [1 - (S + N) / 400]$ KN/m²) Geometry and variable load depend on numerical coefficients corresponding to the third letter of the family name (S) and the third letter of the given name (N). <table><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td><td>L</td><td>M</td><td>N</td><td>O</td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td><td>U</td><td>V</td><td>W</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td><td>21</td><td>22</td><td>23</td><td>24</td><td>25</td><td>26</td></tr></table>	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z																													
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26																													
6	Poisson Ratio	0.18																																																				
7	Unit Weight	25 KN/m ³																																																				

Section 2 Determination of basic unknown's parameter

- **Slab side:**

$$a = 8.5 * [1 + (S + N) / 200] \text{ m}$$

- **Variable load:**

$$q_k = 5.0 * [1 - (S + N) / 400] \text{ KN/m}^2$$

The Variable S and N for **Haseeb Ali** are S=9 and N=19.

Therefore,

$$a = 8.5 * [1 + (9 + 19) / 200] = 9.69 \text{ m} = 969 \text{ cm}$$

$$q_k = 5.0 * [1 - (9 + 19) / 400] = 4.65 \text{ KN/m}^2$$

1. Slab Depth (h):

$$\begin{aligned} h &\geq a/25 \\ 9.69/25 &= 0.3876 \text{ m} \\ &\sim \mathbf{40 \text{ cm}} \end{aligned}$$

2. Loading:

$$\begin{aligned} \text{Self-weight (g}_{0k}) &= \text{Unit wt.} \times \text{slab depth} \\ 25 \times 0.4 &= 10 \text{ KN/m}^2 \\ \text{Non-structural members} &= 1 \text{ KN/m}^2 \\ \mathbf{\text{Total Permanent Load (g}_k)} &= 10 + 1 = 11 \text{ KN/m}^2 \\ \mathbf{\text{Live Load (q}_k)} &= 4.65 \text{ KN/m}^2 \end{aligned}$$

For Ultimate Limit Design the combined load shall be:

As per Euro code 0, Annex A.1.2.1 (2) "The combinations of actions should be used when verifying ultimate limit states". Therefore;

$$\mathbf{\text{Design Loading}} = y_g \times g_k + y_g \times q_k$$

Taking y_g and y_g as 1.35 and 1.5, respectively as per Euro Code Table A1.1 (A)

$$\mathbf{\text{Design Loading (P}_u)} = 1.35 \times 11 + 1.5 \times 4.65 = \mathbf{21.825 \text{ KN/m}^2}$$

3. Mechanical Properties of Materials:

Concrete

Euro code 2, 4.1 (4):

“Corrosion protection of steel reinforcement depends on density, quality and thickness of concrete cover and cracking. The cover density and quality is achieved by controlling the maximum water/cement ratio and minimum cement content and may be related to a minimum strength class of concrete”.

Therefore;

Class = C20/25¹

As per recommendation of Table 3.1 Euro code 2, following values are selected

Characteristic Strength (f_{ck}) = 25 MPa

Design Strength (f_{cd}) = $\alpha_{cc} \times f_{ck} / \gamma$

Taking α_{cc} (long-term impacts) as 0.85 and γ (safety factor) as 1.5

0.85x25/1.5

f_{cd} = **14.17 MPa**

f_{ctm} = **2.2 MPa**

f_{ctm} = **1 MPa**

ϵ_{cu} = **0.0035**

Concrete cover:

Determining concrete cover as per Euro Code 2, 4.4.1:

$$C_{nom} = C_{min} + \Delta C_{dev}$$

$$C_{min} = \max (C_{min,b}; C_{min,dur} + \Delta C_{dur,\gamma} - \Delta C_{dur,st} - \Delta C_{dur,add}; 10\text{mm})$$

$C_{min,b}$ = 16 mm (Table 4.2 Eurocode 2; based on assumed bar size of 16 mm with assumption that nominal maximum aggregate size is less than 32mm)

$C_{min,dur}$ = 15 mm S4; for recommended structural class

$\Delta C_{dur,\gamma}$ = 0 mm Recommended value

$\Delta C_{dur,st}$ = 0 mm Recommended value

$\Delta C_{dur,add}$ = 0 mm Recommended value

$$C_{min} = \max (16; 15; 10) = 16 \text{ mm}$$

$$\Delta C_{dev} = 10\text{mm} (4.4.1.3 (1), \text{Euro code 2})$$

$$C_{nom} = C_{min} + \Delta C_{dev} = 16 + 10 = 26 \text{ mm}$$

Steel (Longitudinal Reinforcement)

Grade =	B450C ²
Characteristic Yielding Strength (f_{yk}) =	450 MPa
Design Yielding Strength (f_{yd}) = f_{yk} / γ_s	
Taking γ_s (safety factor) as 1.15	
	450/1.15
f_{yd} =	391.3 MPa
Modulus of Elasticity (E) =	210 GPa
Yield Strain (ϵ_{yd}) =	0.0018633

Request n.1

Evaluate the bending moments in the elastic linear analysis according to Strip Method. Compare the results with other methods as FEM (linear elastic analysis).

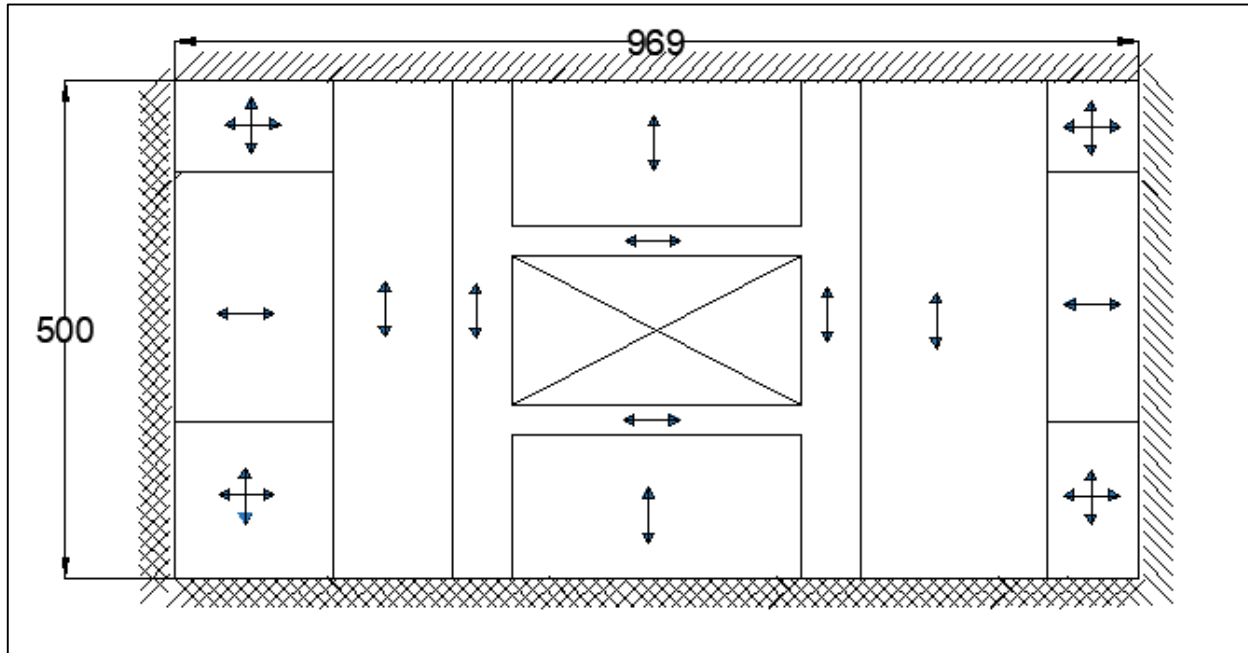


Figure 1: Load Distribution and dimensions of slab

The basic pattern of discontinuity lines and load dispersion will be selected according to Fig1.2 positioned to take into account the greater load carrying capacity of fixed edge than of simply supported edges. The coefficient (β) defines the width of the strips. In the corner load is divided by the coefficient (β). In the centered region, 100 percent of the load is assigned to y-direction except at the opening in strong band. While along the central part of the short edges, 100 percent of the load is carried in x-direction. Moments for this basic case without the hole is calculated and later used as guide in selecting moments for the actual slab with hole. A ratio of support to pan moments of 2.0 will be used generally. Moments for the slab, neglecting the hole.

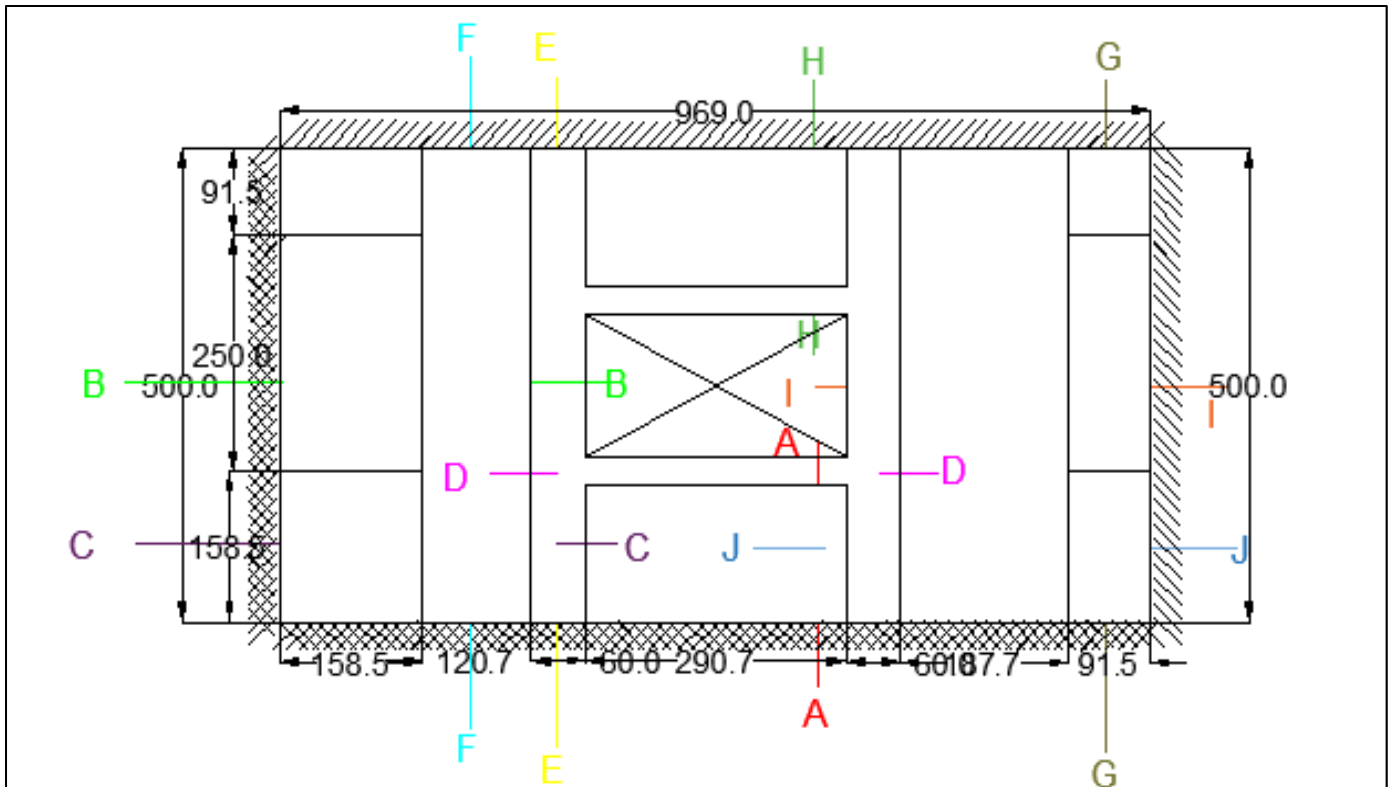


Figure 2: Strips Section of Slab

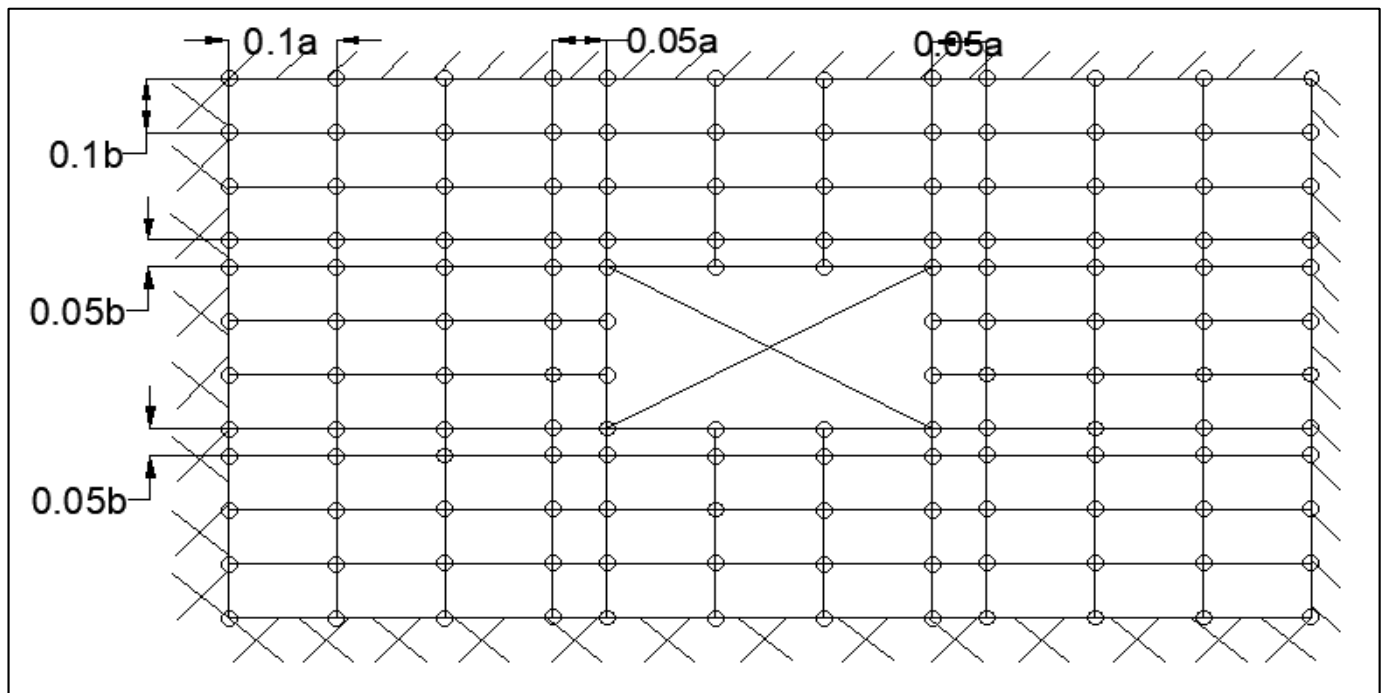


Figure 3: Slab reference grid

4 X-Direction Middle Strip:

The Cantilever moment at the left support is

$$m_x = \frac{(1-\beta)^2 P_u b^2}{8} = 27.41 \frac{KN.m}{m}$$

Since: Ratio of Span to support
Moment is 2, so $\beta=0.366$

So, the negative moment at left support is

$$m_{xs} = \frac{(1-2\beta)P_u b^2}{8} = 18.27 \frac{KN.m}{m}$$

And positive moment at left support is

$$m_{xf} = \frac{m_{xs}}{\frac{1-2\beta}{\beta^2}} = 9.14 \frac{KN.m}{m}$$

The moment in the x-direction edge strip are one half of those in the middle strips because the load is one-half as great.

4 Y-Direction Middle Strip:

The negative moment at left support is

$$m_{ys} = \frac{(1-2\beta)P_u b^2}{2} = 73.11 \frac{KN.m}{m}$$

And positive moment at left support is

$$m_{yf} = \frac{\beta^2 P_u b^2}{2} = 36.54 \frac{KN.m}{m}$$

The moment in the y-direction edge strip are one half of those in the middle strips because the load is one-half as great.

Because of hole, certain strips lack support at one end to support them, 0.3m wide strong band will provided in x-direction at the long edges of the hole and 0.6m wide strong band in the y-direction.

a. Strip A-A:

It may at first be assumed that propped cantilever action is obtained, with the restrained moment along the slab edge taken as $73.11 \frac{KN.m}{m}$, the same as for the basic case. Summing moments about the left end of the loaded strip then result is.

$$w_1 = -104.5 \frac{KN.m}{m}$$

The negative value indicates that the cantilever strip are serving as support for strip D-D, and in turn for the strong bands in the y-direction, which is hardly a reasonable assumption. Instead a discontinuity line will be assumed 1.45m from the support as shown in Fig:3, terminating the cantilever and leaving the 0.3m strip D-D along the edge of the opening in the x-direction to carry its own load. It follows that the support moment in the cantilever strip is negative.

$$m_{ys} = -22.9 \frac{KN.m}{m}$$

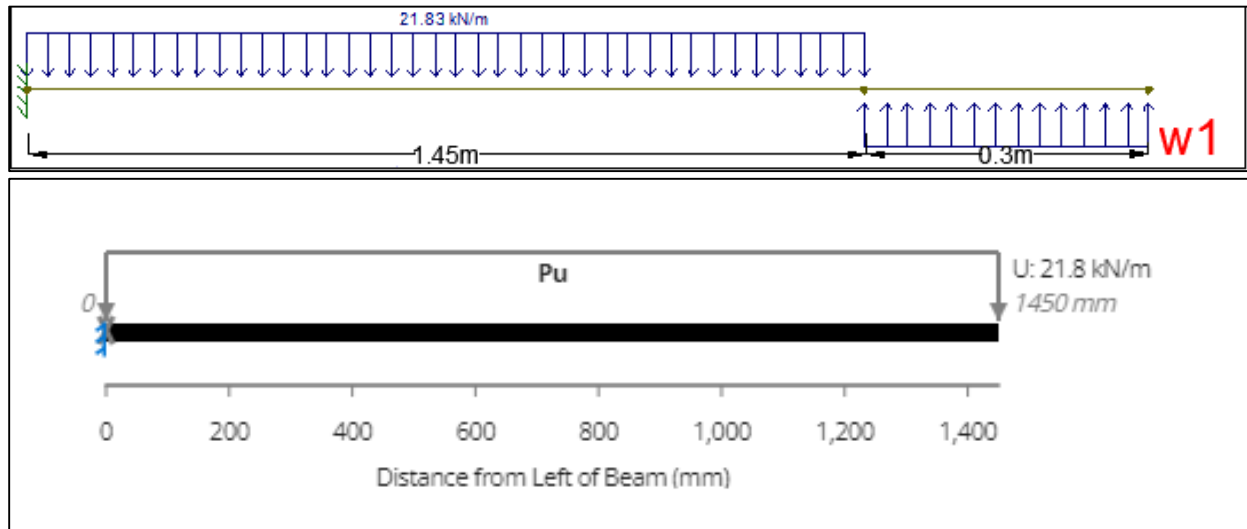


Figure 4: Load Distribution on Strip A-A

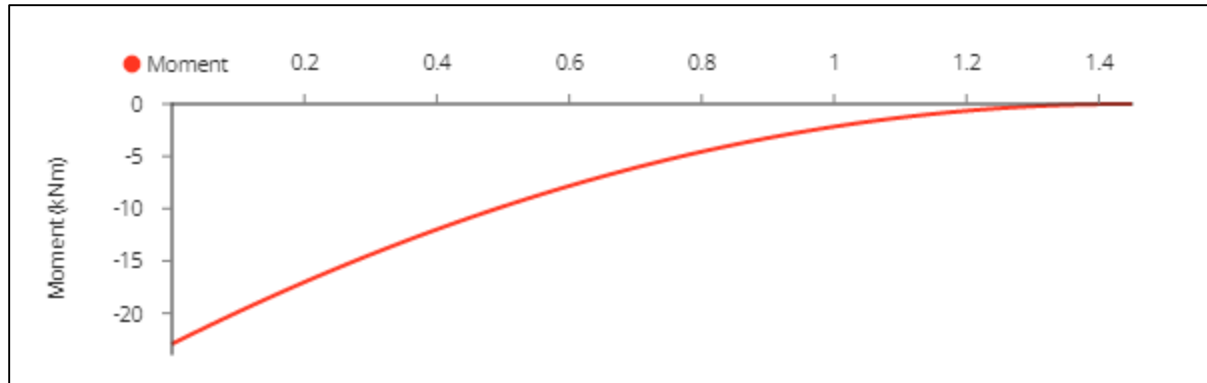


Figure 5: Moment at Strip A-A

b. Strip B-B:

The restraint moment at the supported edge will be taken to be the same as the basic case, i.e. $18.27 \frac{KN.m}{m}$. Summing moments about the left end of the strip of Fig 5, then result in an uplift reaction at the right end, to be provided by strip E-E,

$$w_2 = -4.84 \frac{KN}{m^2}$$

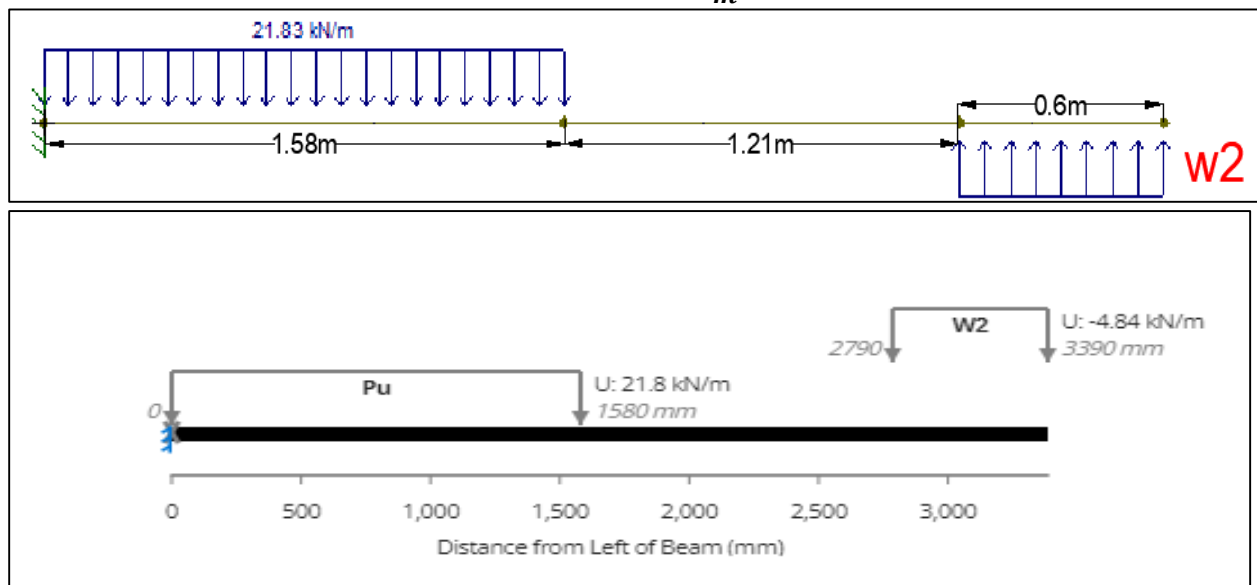


Figure 6: Load Distribution on Strip B-B

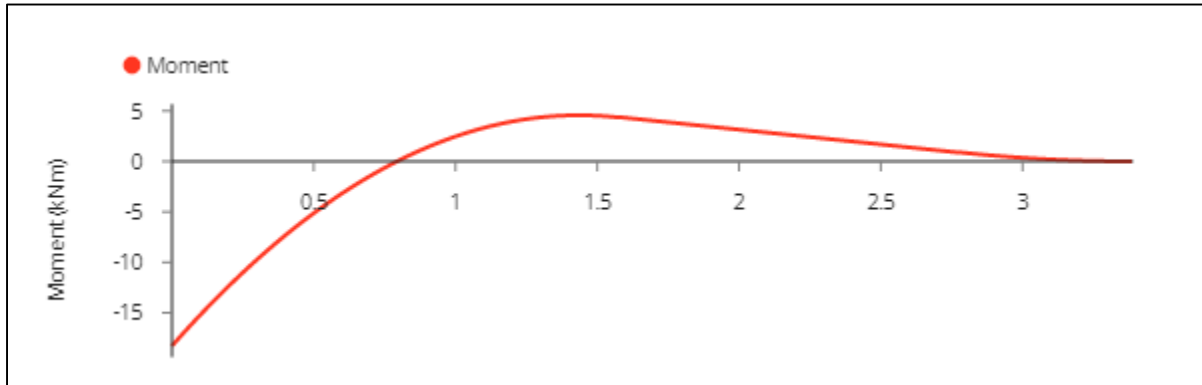


Figure 7: Moment at Strip B-B

$$m_{xs} = -18.3 \frac{KN.m}{m}$$

$$m_{xf} = 4.57 \frac{KN.m}{m} @ 1.43m$$

c. Strip C-C:

Negative and positive moments and reactions to be provided by strip E-E as shown in Fig 7, are all one-half the corresponding values for strip C-C.

$$w_3 = -2.42 \frac{KN}{m^2}$$

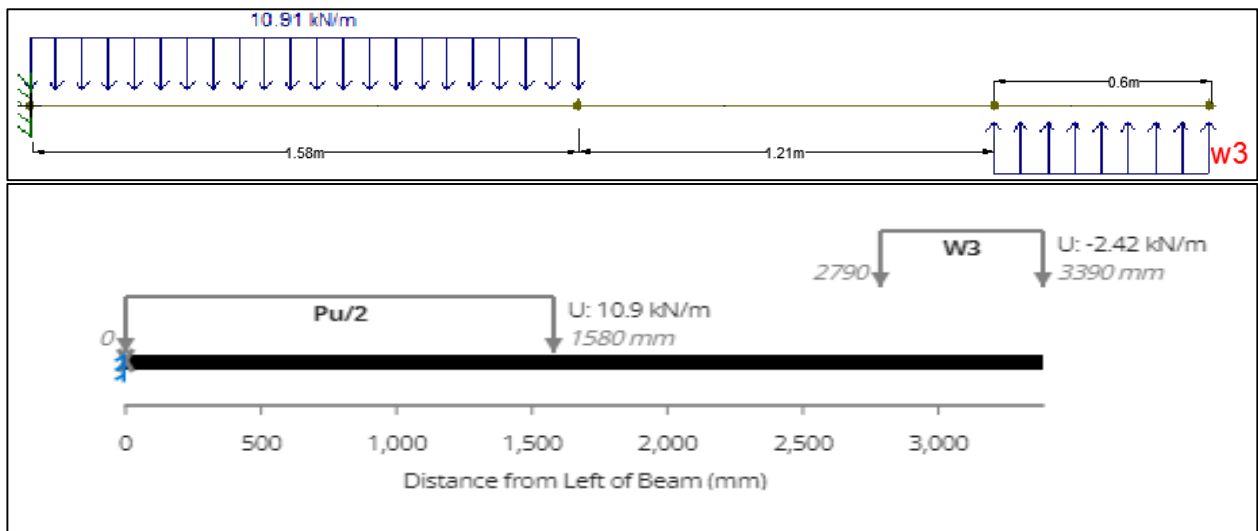


Figure 8: Load Distribution on Strip C-C

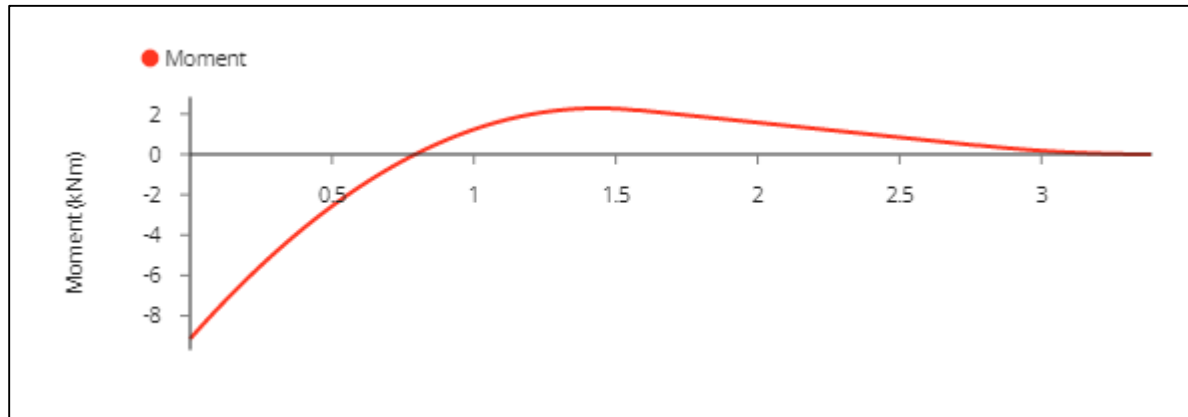


Figure 9: Moment at Strip C-C

$$m_{xs} = -9.14 \frac{KN.m}{m}$$

$$m_{xf} = 2.28 \frac{KN.m}{m} @ 1.45m$$

d. Strip D-D:

The 0.3m wide strip carries $21.825 \frac{KN}{m^2}$ in the x-direction with the reactions provided by the strong bands E-E as show in Fig 9.

$$w_4 = -52.87 \frac{KN}{m^2}$$

$$m_{xf} = 32.6 \frac{KN.m}{m} @ 2.06m$$

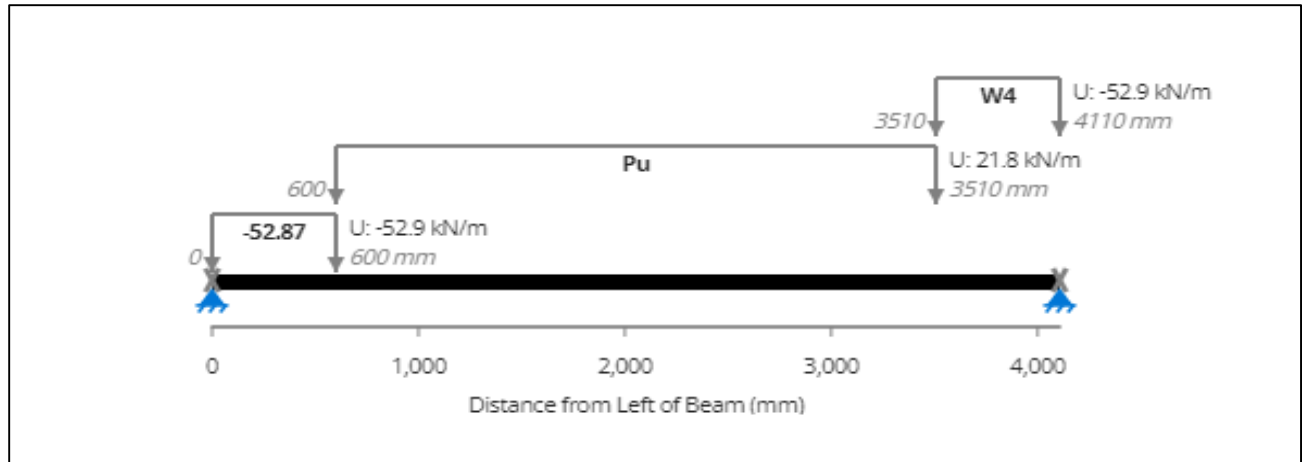


Figure 10: Load Distribution on Strip D-D

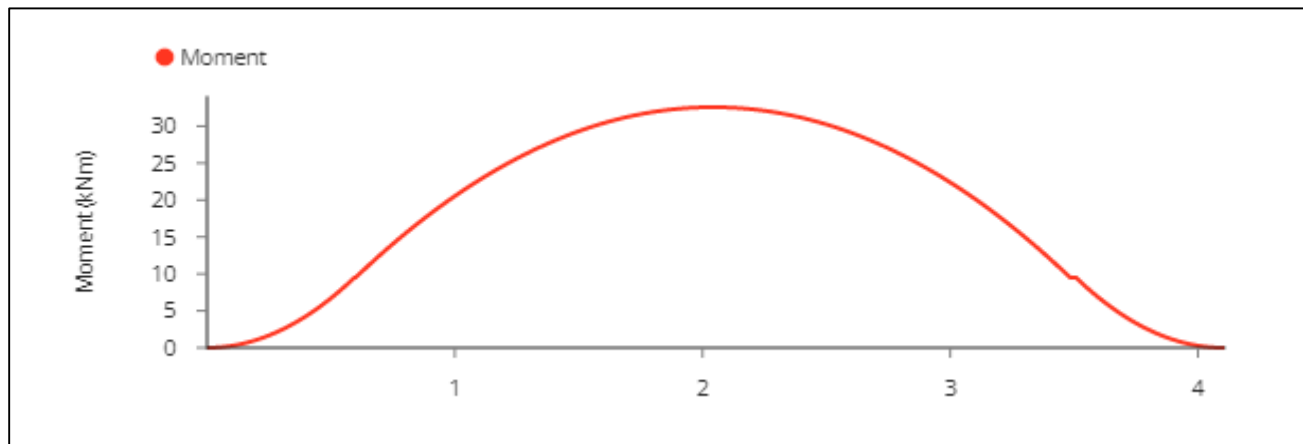


Figure 11: Moment at Strip D-D

e. Strip E-E:

In reference Fig 11, the strong band in the y-direction carry the directly load of $21.825 \frac{KN}{m^2}$ plus the $4.84 \frac{KN}{m^2}$ load from the strip B-B, the $2.42 \frac{KN}{m^2}$ load from strip C-C, and the $52.87 \frac{KN}{m^2}$ end reaction from strip D-D, So for Strip E-E.

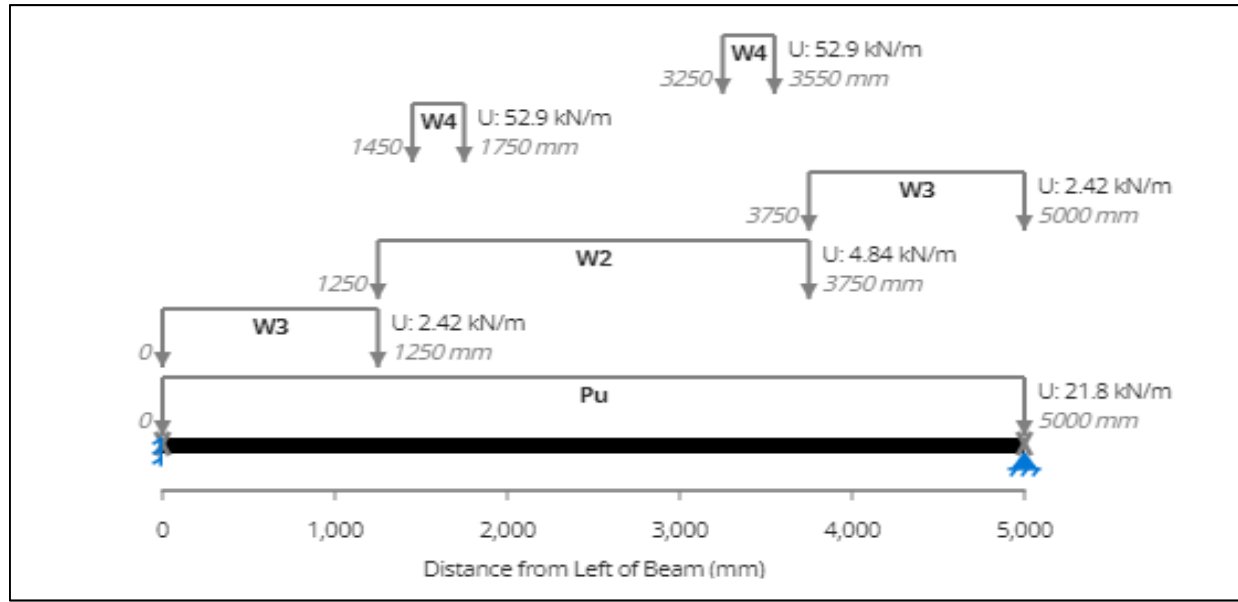


Figure 12: Load Distribution on Strip E-E

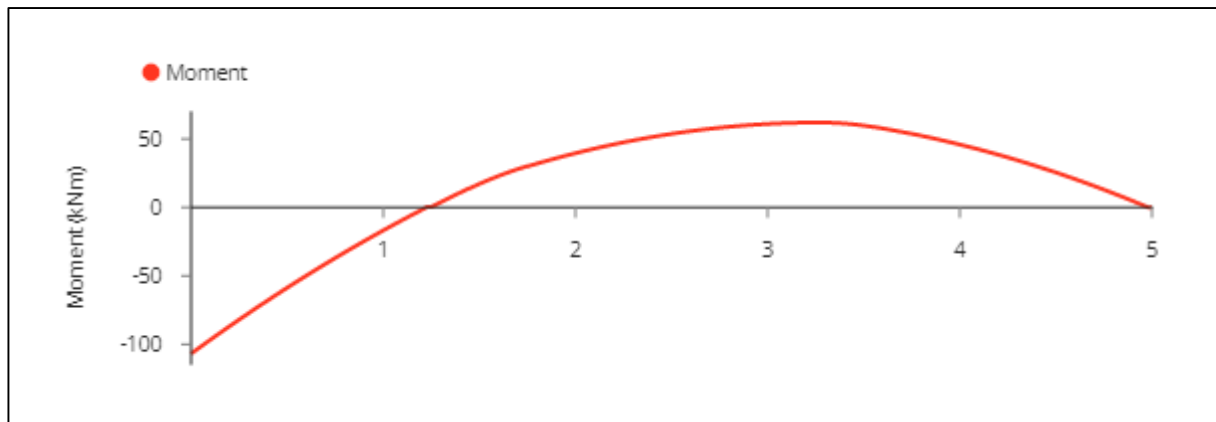


Figure 13: Moment at Strip E-E

$$m_{ys} = -107 \frac{\text{KN.m}}{\text{m}}$$

$$m_{yf} = 61.9 \frac{\text{KN.m}}{\text{m}} @ 3.3\text{m}$$

f. Strip F-F:

The moment for the y-direction middle strip of the basic case may be used without change thus, Fig 12.

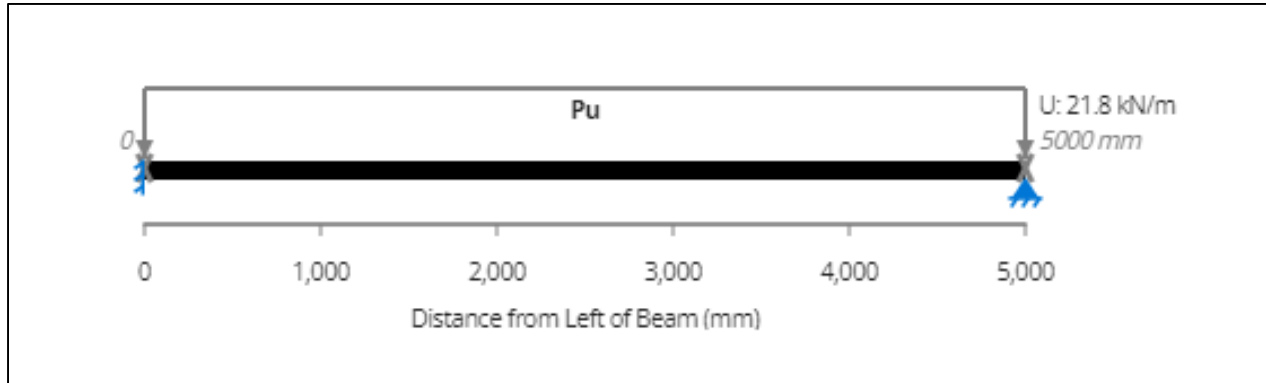


Figure 14: Load Distribution on Strip F-F

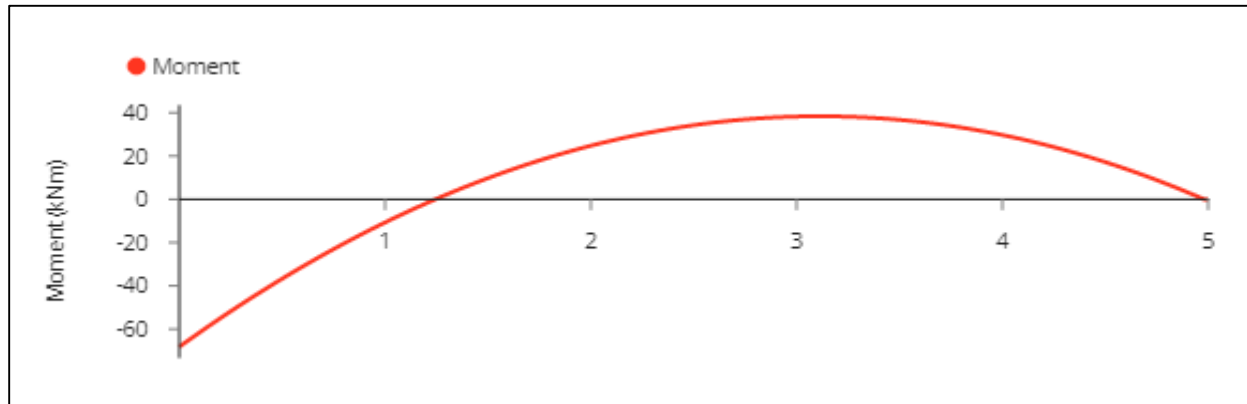


Figure 15: Moment at Strip F-F

$$m_{ys} = -68.2 \frac{KN.m}{m}$$

$$m_{yf} = 38.4 \frac{KN.m}{m} @ 3.1m$$

g. Strip G-G:

The moment for the y-direction middle strip of the basic case may be used without change thus, Fig 13.

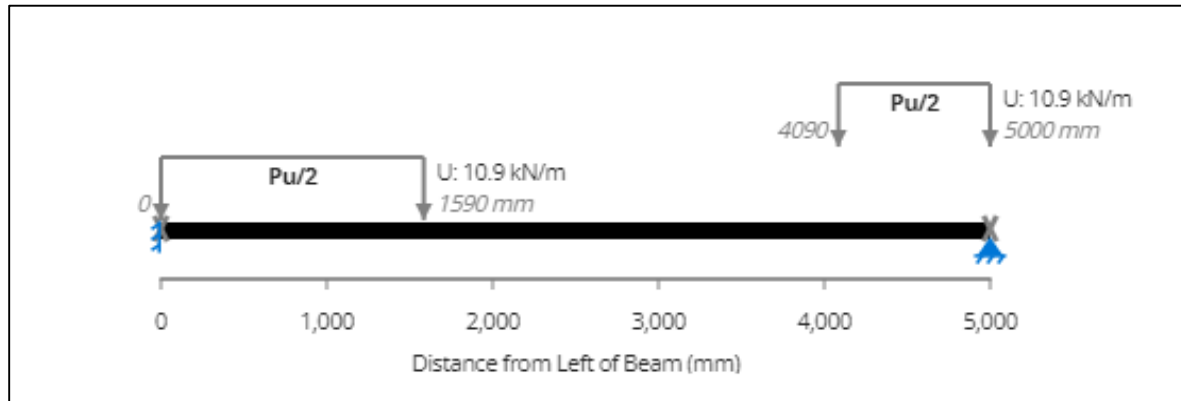


Figure 16: Load Distribution on Strip G-G

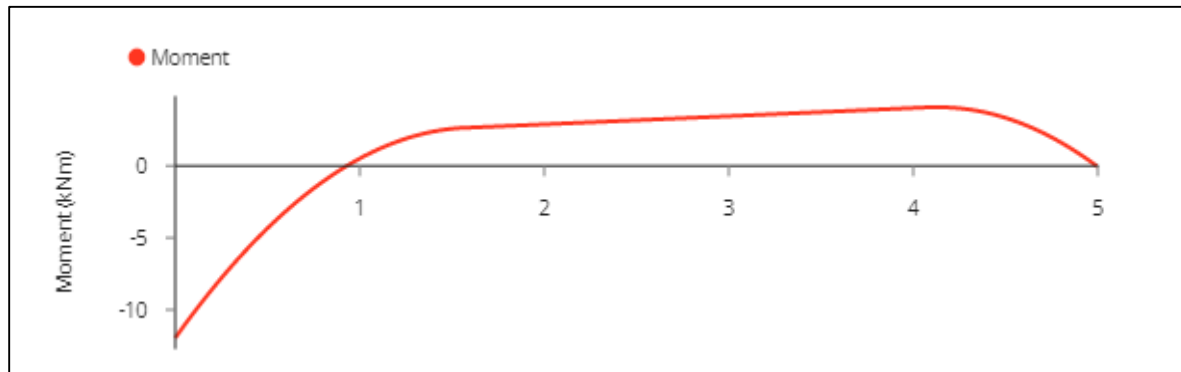


Figure 17: Moment at Strip G-G

$$m_{ys} = -12 \frac{KN.m}{m}$$

$$m_{yf} = 4.07 \frac{KN.m}{m} @ 4.15m$$

h. Strip H-H:

The moment for the y-direction middle strip of the basic case may be used without change thus, Fig 14.

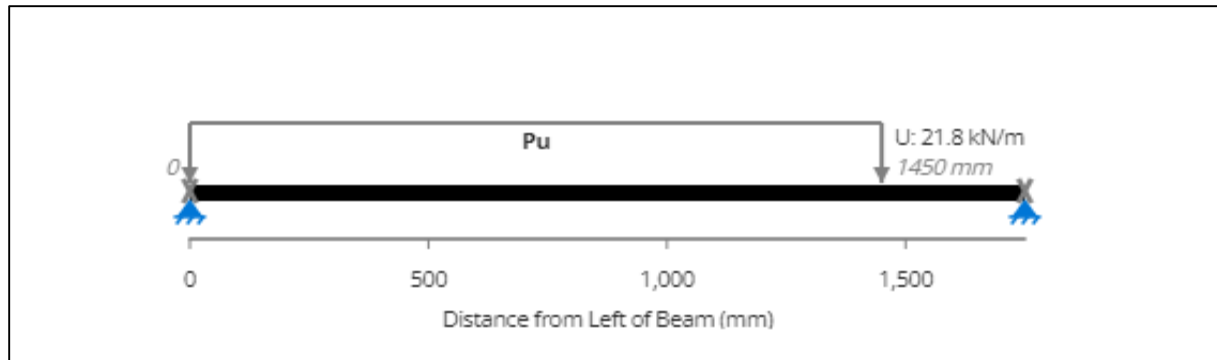


Figure 18: Load Distribution on Strip H-H

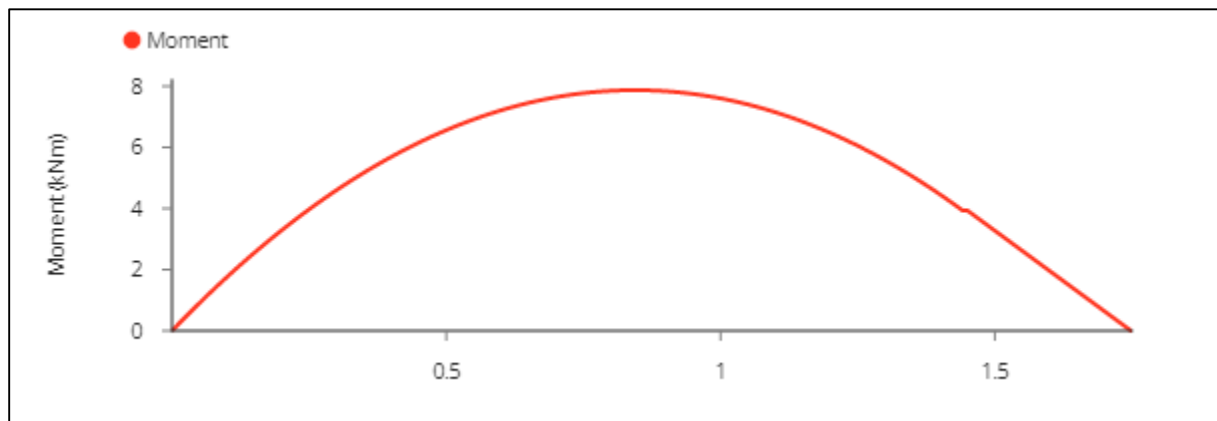


Figure 19: Moment at Strip H-H

$$m_{yf} = 7.87 \frac{KN.m}{m} @ 0.87m$$

i. Strip I-I:

The moment for the x-direction middle strip of the basic case may be used without change thus, Fig 15.

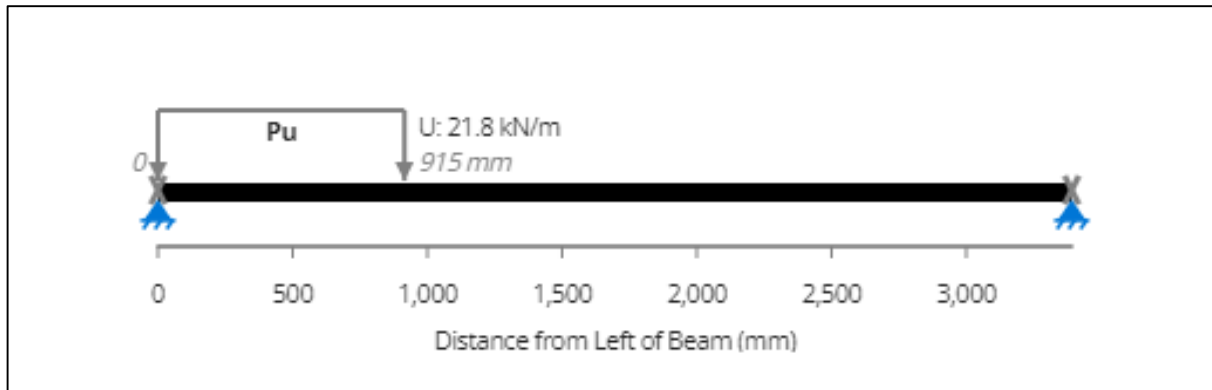


Figure 20: Load Distribution on Strip I-I

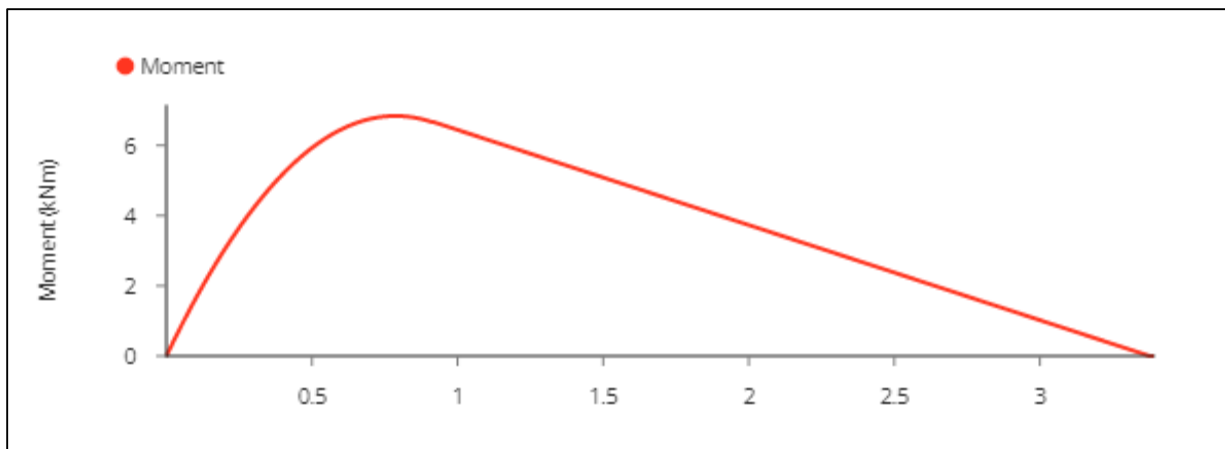


Figure 21: Moment at Strip I-I

$$m_{xf} = 6.84 \frac{KN.m}{m} @ 0.8m$$

j. **Strip J-J:**

The moment for the x-direction middle strip of the basic case may be used without change thus, Fig 16.

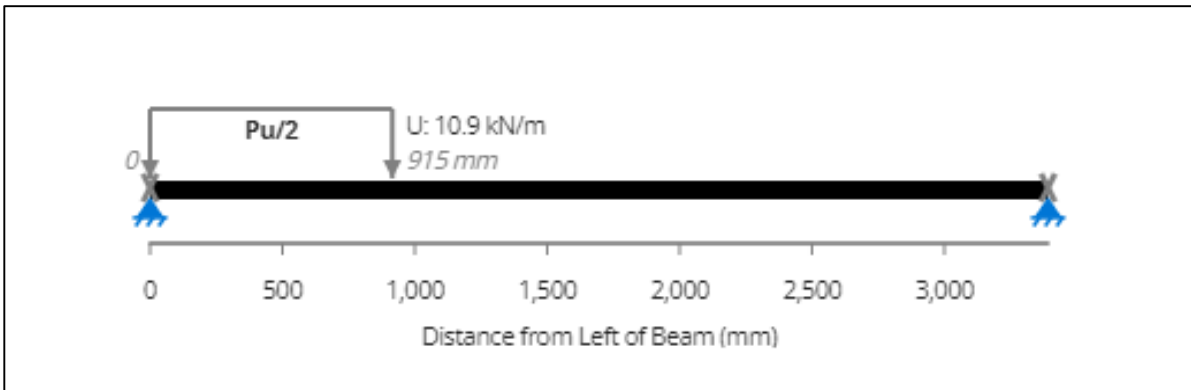


Figure 22: Load Distribution on Strip J-J

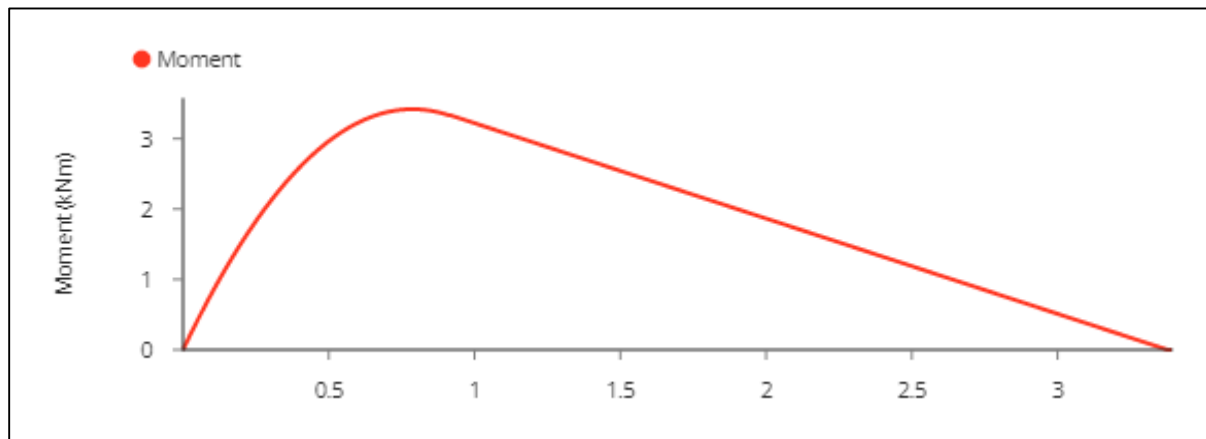


Figure 23: Moment at Strip J-J

$$m_{xf} = 3.42 \frac{KN.m}{m} @ 0.8m$$

Finite Element Method

The finite element method is an approximate technique. The number of elements used in a model can greatly affect the accuracy of the solution. In general, as the number of elements, or the fineness of the mesh, is increased, the accuracy of the model increases as well. As multiple models are created with an increasingly finer mesh, the results should converge to the correct numerical solution such that a significant increase in the number of elements produces an insignificant change in a particular response quantity. Not all response quantities will converge at the same rate, however. Displacements will generally be the most accurate response quantity computed and will converge faster than stresses. Generally, however, convergence of displacements does not guarantee convergence of stresses since stresses are computed as derivatives of the displacement field. Furthermore, it may be necessary to increase the number of elements in areas of the model near either concentrated loads or boundary conditions, where the stress gradient is steeper. The type of element applied in the analysis also can significantly affect the quality of the results because various finite elements are derived from different assumptions. The sum of reactions of the model should be verified with hand calculations. The magnitude of global displacements should be checked and verified to be small with respect to the geometry of the structure. The equilibrium of the structure must be verified, as the finite element method does not guarantee that this important condition is satisfied. The designer must determine whether or not the results make physical sense. Contour plots of stresses and the displaced shape of the structure should roughly correspond to the intuitive sense of the designer. For example, symmetric results must accompany a structure with symmetric geometry and symmetric loading.

Any deviation between expected and computed results must be critically investigated

Computation of BM by a FEM commercial software in the case we use Straus 7 software to analyses the slab. The slab is divided in to 1056 elements and 1144 nodes. One loading case applied to the FE analysis which is the factored load 21.8255 KN/m^2 . The boundary condition and geometry of the slab are defined to the code as prescribed in the problem statement. The outcome result is show in the following figures.

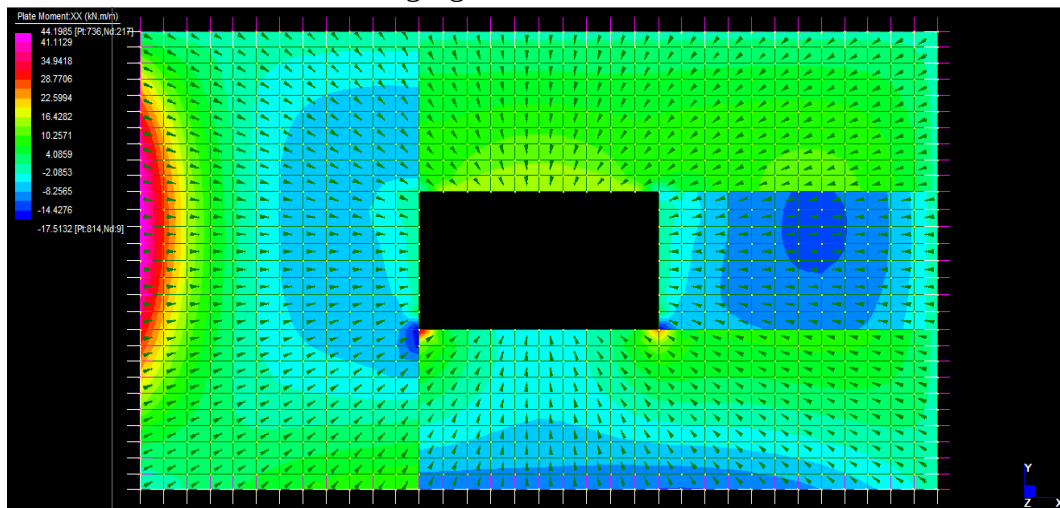


Figure 24: Moment at XX plane

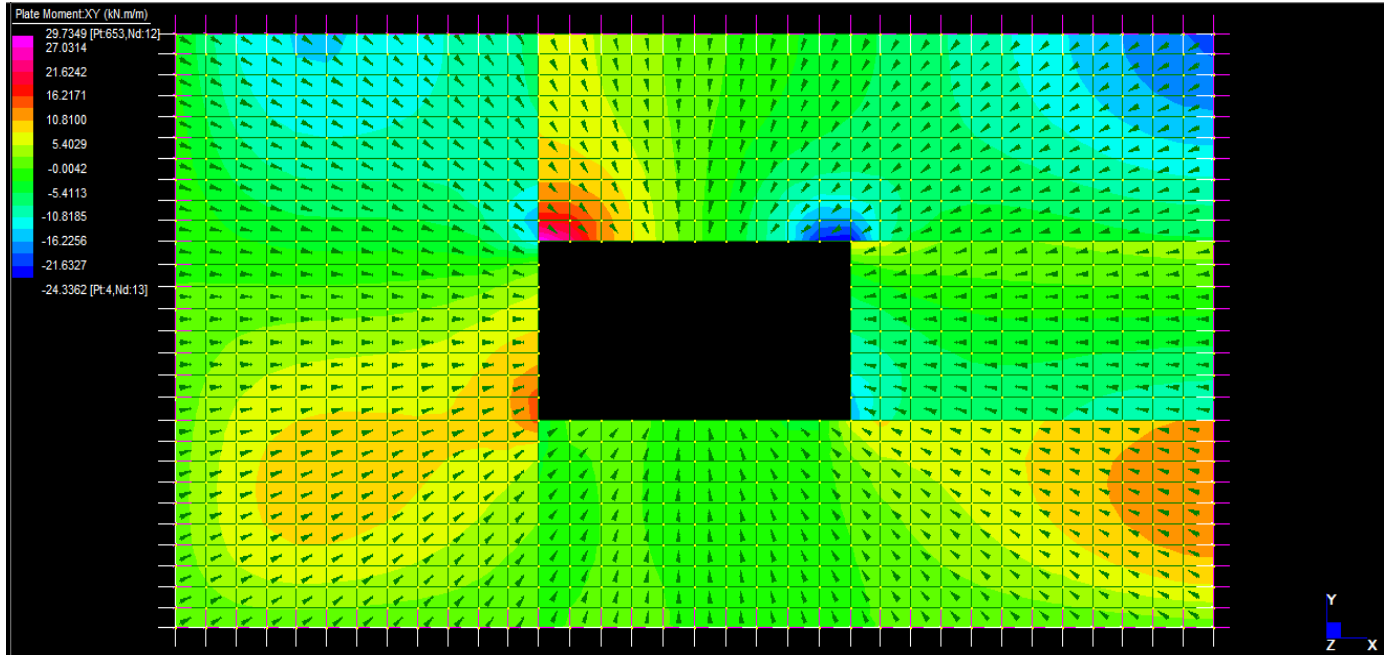


Figure 25: Moment at YY plane

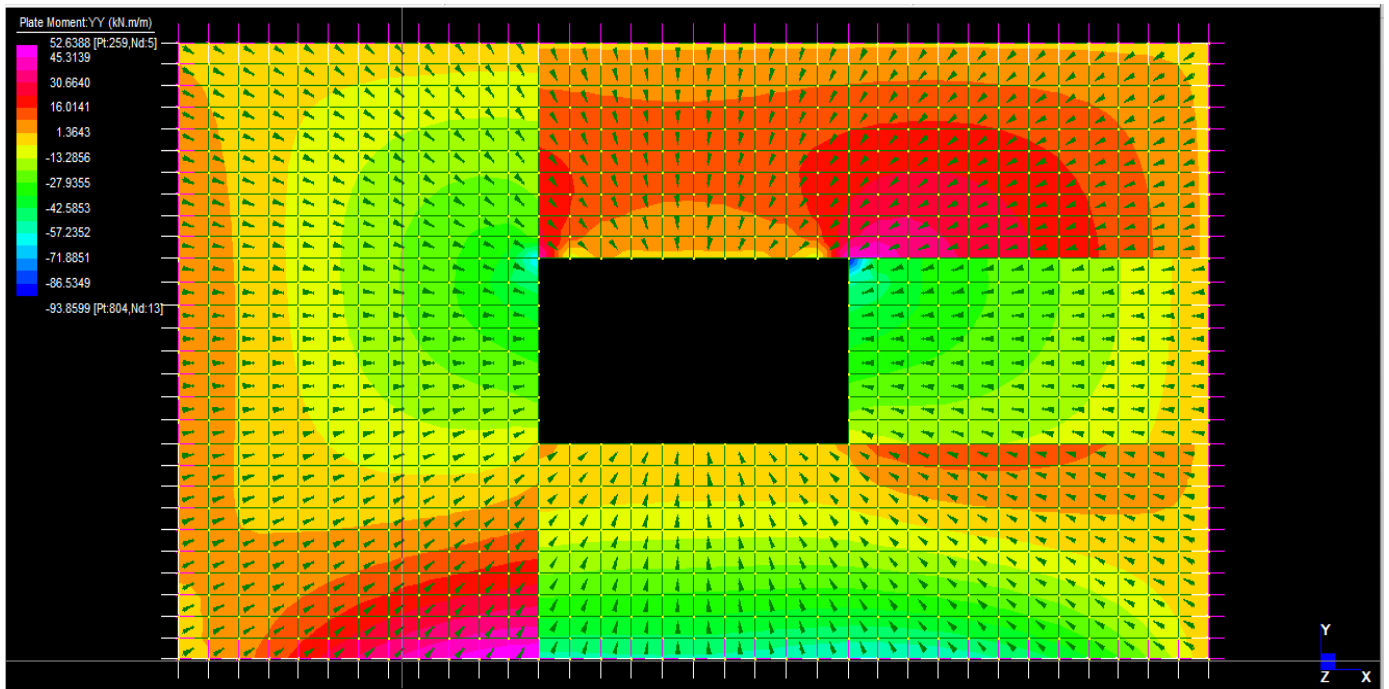


Figure 26: Moment at XY plane

Comparison of B.M, evaluated in both strip method and FEM.

In the sections of this paper bending moments in different points on the slab evaluated both in simple and strong band approaches by strip method, the figures illustrate the location and values of the bending moments, as well as in the section 3.1. The bending moment evaluated by FEM. Comparing the values of these moments in the critical points, both method have same approach in the case of critical sections and variation of the moment along both direction (x, y). since the strip method is a lower bound approach in the limit analysis, in some points the moments are higher than the values found by FEM, in particular at the location of strong bands, bending moments are significant higher than evaluate by FEM. However, in the FEM moments are calculated by computer and the method known as an accurate analysis method therefore we consider the bending moment's outcome from FEM for design of reinforcement in the Wood Aremer method.

Request n.2

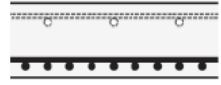
According to Wood-Armer method, design the steel reinforcement and develop technical drawings with reinforcement detailing. Comment on the effectiveness of the reinforcement layouts proposed by comparing resisting and acting bending moments.

Since the Wood-Armer method has local design approach whereby applied moments are required to be determined at each point of slab (as per given resolution of 0.1ax0.1b). The equations referred for the determination of the applied moments for application of this methods are given below:

Bottom Reinforcement

$$\begin{cases} m_{xu}^+ = M_{xu} + |M_{xyu}| = \mathcal{M}_{xu}^+ \\ m_{yu}^+ = M_{yu} + |M_{xyu}| = \mathcal{M}_{yu}^+ \end{cases} \text{ if } \mathcal{M}_{xu}^+ \geq 0 \text{ and } \mathcal{M}_{yu}^+ \geq 0$$

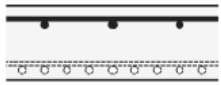
$$\begin{cases} m_{xu}^+ = M_{xu} + \frac{M_{xyu}^2}{|M_{yu}|} \geq 0 \\ m_{yu}^+ = 0 \end{cases} \text{ if } \mathcal{M}_{yu}^+ < 0$$

$$\begin{cases} m_{xu}^+ = 0 \\ m_{yu}^+ = M_{yu} + \frac{M_{xyu}^2}{|M_{xu}|} \geq 0 \end{cases} \text{ if } \mathcal{M}_{xu}^+ < 0$$


Top Reinforcement

$$\begin{cases} m_{xu}^- = M_{xu} - |M_{xyu}| = \mathcal{M}_{xu}^- \\ m_{yu}^- = M_{yu} - |M_{xyu}| = \mathcal{M}_{yu}^- \end{cases} \text{ if } \mathcal{M}_{xu}^- \leq 0 \text{ and } \mathcal{M}_{yu}^- \leq 0$$

$$\begin{cases} m_{xu}^- = M_{xu} - \frac{M_{xyu}^2}{|M_{yu}|} \leq 0 \\ m_{yu}^- = 0 \end{cases} \text{ if } \mathcal{M}_{yu}^- > 0$$

$$\begin{cases} m_{xu}^- = 0 \\ m_{yu}^- = M_{yu} - \frac{M_{xyu}^2}{|M_{xu}|} \leq 0 \end{cases} \text{ if } \mathcal{M}_{xu}^- > 0$$


The moments determined are used as input in the above equations in MS Excel file where algorithm was developed to determine the applied moments. The resultant applied moments are given in the following tables:

M_x^+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	9	12	9	9	9	16	10	15	15	15	15	21	32
0.1	18	12	5	5	9	15	15	15	15	15	15	21	32
0.2	34	6	0	0	0	11	15	15	15	15	15	21	21
0.3	40	6	0	0	0	14	12	15	27	15	11	11	16
0.35	40	6	0	0	0	27	16	15	32	15	11	11	16
0.4	40	6	0	0	0					3	0	0	9
0.5	40	6	0	0	0					3	0	0	9
0.6	40	6	0	0	0					14	0	0	9
0.65	34	6	0	0	3					26	5	5	9
0.7	18	12	5	5	0	9	2	2	15	9	9	9	9
0.8	18	6	3	9	15	0	0	0	0	9	10	15	15
0.9	9	6	3	3	9	0	0	0	0	0	0	1	8
1.00	9	6	3	3	9	0	0	0	0	0	0	0	2

Table 1: Bending Moments (Positive in x-direction)

M_y^+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	7	5	5	12	12	7	1	7	7	7	7	12	23
0.1	7	5	5	5	5	0	21	13	13	7	7	12	23
0.2	7	0	0	0	0	0	21	13	13	13	13	19	12
0.3	7	0	0	0	0	0	17	13	24	13	0	0	7
0.35	7	0	0	0	0	0	21	13	30	13	0	0	7
0.4	7	0	0	0	0					13	0	0	7
0.5	7	0	0	0	0					13	0	0	7
0.6	7	0	0	0	0					24	0	0	7
0.65	7	7	0	0	7					23	0	0	7
0.7	7	12	5	5	7	21	7	7	7	7	21	21	7
0.8	7	7	7	5	41	21	7	7	7	7	7	12	12
0.9	7	14	21	36	36	36	0	0	0	0	0	0	12
1.00	7	14	36	50	50	50	0	0	0	0	0	0	7

Table 2: Bending Moments (Positive in y-direction)

M_x^-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	-1	-10	-13	-13	-13	0	0	0	0	0	0	-1	-11
0.1	0	-10	-17	-17	-13	0	0	0	0	0	0	-1	-11
0.2	0	-4	-14	-14	-14	0	0	0	0	0	0	-1	-1
0.3	0	-4	-14	-14	-14	-1	0	0	-6	0	0	0	0
0.35	0	-4	-14	-16	-10	-11	0	0	-11	0	0	0	0
0.4	0	-4	-14	-16	-10					-5	-7	-20	-1
0.5	0	-4	-14	-14	-7					-5	-7	-20	-1
0.6	0	-4	-14	-14	-7					-18	-7	-20	-1
0.65	0	-4	-14	-14	-7					-18	-1	-1	-1
0.7	0	-10	-17	-17	-25	0	-8	-8	0	-1	0	0	-1
0.8	0	-4	-7	-13	0	-9	-13	-13	-13	-1	-1	-7	-7
0.9	-1	-2	-4	-3	0	-9	-13	-13	-13	-8	-6	-14	-14
1.00	-1	-2	-3	-3	0	-13	-18	-18	-18	-18	-18	-9	-8

Table 3: Bending Moments (negative in x-direction)

M_y^-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	-4	-17	-17	-9	-9	-1	0	-1	-1	-1	-1	-9	-20
0.1	-1	-17	-17	-17	-17	-9	0	0	0	-1	-1	-9	-20
0.2	0	-12	-16	-33	-33	-31	0	0	0	0	0	-3	-9
0.3	0	-12	-16	-33	-33	-39	0	0	-8	0	-31	-31	-1
0.35	0	-12	-16	-36	-36	-50	0	0	-13	0	-31	-31	-1
0.4	0	-12	-16	-36	-36					0	-33	-33	-4
0.5	0	-12	-16	-33	-33					0	-33	-33	-4
0.6	0	-12	-16	-33	-33					-8	-33	-33	-4
0.65	0	-4	-16	-33	-4					-20	-33	-33	-4
0.7	-1	-9	-17	-17	-9	0	-4	-4	-1	-4	0	0	-4
0.8	-1	-4	-4	-17	0	0	-4	-4	-4	-4	-4	-9	-9
0.9	-4	0	0	0	0	0	-33	-33	-33	-33	-31	-39	-9
1.00	-4	0	0	0	0	0	-33	-33	-33	-33	-33	-33	-4

Table 4: Bending Moments (negative in y-direction)

The rebar diameter has fixed as $\varnothing 16$ [mm] for the bottom reinforcements, while for the top the bar diameter of $\varnothing 12$ [mm] is selected. The first step is the computation of the amount of reinforcement. Knowing the bending moment (obtained by Wood-Armer Method), by mean of the pre-dimensioning formula we are able to obtain the theoretical amount of reinforcement.

Pre-Dimensioning Formula:

$$A_s = m / (0.9 * b * d * f_{yd})$$

Where:

m = applied bending moment (from above tables)

b = 1 meter (unit section width)

d =

For x^+ ; d = slab depth – cover – bar diameter/2 = $400 - 26 - 16/2 = 366$ mm

For y^+ ; d = slab depth – cover – bar diameter (x^+) - bar diameter/2 = $400 - 26 - 16 - 16/2 = 350$ mm

For x^- ; d = slab depth – cover – bar diameter (y^-) - bar diameter/2 = $400 - 26 - 12 - 12/2 = 356$ mm

For y^- ; d = slab depth – cover – bar diameter/2 = $400 - 26 - 12/2 = 368$ mm

f_{yd} = design value of yielding stress for the steel

A_s = Required steel from pre-dimensioning formula

Based on this, the area of required steel in mm^2 is:

A_{sx}^+ (mm^2)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	74	92	68	68	68	121	80	119	119	119	119	164	247
0.1	138	92	35	35	68	116	119	119	119	119	119	164	247
0.2	265	45	0	0	0	88	119	119	119	119	119	164	164
0.3	313	45	0	0	0	112	90	119	206	119	87	87	122
0.35	313	45	0	0	2	209	121	119	247	119	87	87	122
0.4	313	45	0	0	2					24	0	0	74
0.5	313	45	0	0	0					24	0	0	74
0.6	313	45	0	0	0					110	0	0	74
0.65	265	50	0	0	26					200	39	39	74
0.7	138	92	35	35	0	72	18	18	119	72	72	72	74
0.8	138	50	26	68	116	0	0	0	0	72	74	116	116
0.9	74	48	24	26	74	0	0	0	0	0	0	9	60
1.00	74	48	24	24	72	0	0	0	0	0	0	0	18

Table 5: Required bottom steel in x-direction

Asy^+ (mm ²)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	55	37	37	99	99	55	11	53	53	53	53	99	187
0.1	55	37	37	37	37	0	172	108	108	53	53	99	187
0.2	55	0	0	0	0	0	172	108	108	108	108	154	99
0.3	55	0	0	0	0	0	140	108	198	108	0	0	55
0.35	55	0	0	0	0	0	174	108	242	108	0	0	55
0.4	55	0	0	0	0					108	0	0	55
0.5	55	0	0	0	0					108	0	0	55
0.6	55	0	0	0	0					198	0	0	55
0.65	55	55	0	0	55					187	0	0	55
0.7	55	99	37	37	55	172	53	53	53	53	172	172	55
0.8	55	55	55	37	336	172	53	53	53	53	55	99	99
0.9	55	114	172	293	293	293	0	0	0	0	0	0	99
1.00	55	114	290	409	409	409	0	0	0	0	0	0	55

Table 6: Required bottom steel in y-direction

Asx^- (mm ²)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	11	78	103	103	103	0	0	0	0	0	0	4	91
0.1	0	78	136	136	103	0	0	0	0	0	0	4	91
0.2	0	35	109	109	109	0	0	0	0	0	0	4	4
0.3	0	35	109	109	109	4	0	0	48	0	0	0	0
0.35	0	35	109	131	82	91	0	0	91	0	0	0	0
0.4	0	35	109	131	82					43	58	156	11
0.5	0	35	109	109	60					43	58	156	11
0.6	0	35	109	109	60					146	58	156	11
0.65	0	34	109	109	60					140	8	8	11
0.7	0	78	136	136	201	0	64	64	0	8	0	0	11
0.8	0	34	60	103	0	73	107	107	107	8	11	54	54
0.9	11	15	30	24	0	73	107	107	107	66	47	111	111
1.00	11	15	24	21	0	104	140	140	140	140	140	68	68

Table 7: Required top steel in x-direction

As_y^- (mm ²)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	31	132	132	73	73	11	0	9	9	9	9	73	156
0.1	8	132	132	132	132	70	0	0	0	9	9	73	156
0.2	0	90	126	257	257	238	0	0	0	0	0	21	73
0.3	0	90	126	257	257	299	0	0	63	0	236	236	12
0.35	0	90	126	279	279	383	0	0	104	0	236	236	12
0.4	0	90	126	279	279					0	255	255	31
0.5	0	90	126	257	257					0	255	255	31
0.6	0	90	126	257	257					62	255	255	31
0.65	0	31	126	257	31					156	255	255	31
0.7	8	73	132	132	73	0	29	29	9	29	0	0	31
0.8	8	31	31	132	0	0	29	29	29	29	32	73	73
0.9	31	0	0	0	0	0	255	255	255	255	237	299	73
1.00	31	0	0	0	0	0	255	255	255	255	255	257	31

Table 8: Required top steel in y-direction

Based on the results, the steel reinforcement provided in the slab is given below. For the reinforcement in +ve direction, Ø16 bar is used whereas for -ve reinforcement Ø12 bar is used. The design steel reinforcement is given below:

Bars Spacing (x+)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	30	30	30	30	30	30	30	30	30	30	30	30	30
0.1	30	30	30	30	30	30	30	30	30	30	30	30	30
0.2	30	30	30	30	30	30	30	30	30	30	30	30	30
0.3	30	30	30	30	30	30	30	30	30	30	30	30	30
0.35	30	30	30	30	30	30	30	30	30	30	30	30	30
0.4	30	30	30	30	30					30	30	30	30
0.5	30	30	30	30	30					30	30	30	30
0.6	30	30	30	30	30					30	30	30	30
0.65	30	30	30	30	30					30	30	30	30
0.7	30	30	30	30	30	30	30	30	30	30	30	30	30
0.8	30	30	30	30	30	30	30	30	30	30	30	30	30
0.9	30	30	30	30	30	30	30	30	30	30	30	30	30
1.00	30	30	30	30	30	30	30	30	30	30	30	30	30

Table 9: Bottom bar (Ø16) spacing selected for design x-direction

Bars Spacing (y+)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	30	30	30	30	30	30	30	30	30	30	30	30	30
0.1	30	30	30	30	30	30	30	30	30	30	30	30	30
0.2	30	30	30	30	30	30	30	30	30	30	30	30	30
0.3	30	30	30	30	30	30	30	30	30	30	30	30	30
0.35	30	30	30	30	30	30	30	30	30	30	30	30	30
0.4	30	30	30	30	30					30	30	30	30
0.5	30	30	30	30	30					30	30	30	30
0.6	30	30	30	30	30					30	30	30	30
0.65	30	30	30	30	30					30	30	30	30
0.7	30	30	30	30	30	30	30	30	30	30	30	30	30
0.8	30	30	30	30	30	30	30	30	30	30	30	30	30
0.9	30	30	30	30	30	30	30	30	30	30	30	30	30
1.00	30	30	30	30	30	30	30	30	30	30	30	30	30

Table 10: Bottom bar (Ø16) spacing selected for design y-direction

Bars Spacing (x-)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	25	25	25	25	25	25	25	25	25	25	25	25	25
0.1	25	25	25	25	25	25	25	25	25	25	25	25	25
0.2	25	25	25	25	25	25	25	25	25	25	25	25	25
0.3	25	25	25	25	25	25	25	25	25	25	25	25	25
0.35	25	25	25	25	25					25	25	25	25
0.4	25	25	25	25	25					25	25	25	25
0.5	25	25	25	25	25					25	25	25	25
0.6	25	25	25	25	25					25	25	25	25
0.65	25	25	25	25	25	25	25	25	25	25	25	25	25
0.7	25	25	25	25	25	25	25	25	25	25	25	25	25
0.8	25	25	25	25	25	25	25	25	25	25	25	25	25
0.9	25	25	25	25	25	25	25	25	25	25	25	25	25
1.00	25	25	25	25	25	25	25	25	25	25	25	25	25

Table 11: Top bar (Ø12) spacing selected for design x-direction

Bars Spacing (y-)	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	25	25	25	25	25	25	25	25	25	25	25	25	25
0.1	25	25	25	25	25	25	25	25	25	25	25	25	25
0.2	25	25	25	20	20	20	25	25	25	25	25	25	25
0.3	25	25	25	20	20	15	25	25	25	25	20	20	25
0.35	25	25	25	20	20					25	20	20	25
0.4	25	25	25	20	20					25	20	20	25
0.5	25	25	25	20	20					25	20	20	25
0.6	25	25	25	20	20					25	20	20	25
0.65	25	25	25	20	25	25	25	25	25	25	20	20	25
0.7	25	25	25	25	25	25	25	25	25	25	25	25	25
0.8	25	25	25	25	25	25	25	25	25	25	25	25	25
0.9	25	25	25	25	25	25	20	20	20	20	20	20	25
1.00	25	25	25	25	25	25	20	20	20	20	20	20	25

Table 12: Bottom bar (Ø12) spacing selected for design y-direction

Minimum and maximum amount of reinforcement:

Euro code 2 – 7.3.1(P):

”Cracking shall be limited to an extent that will not impair the proper functioning or durability of the structure or cause its appearance to be unacceptable”.

Euro code 2 - 9.2.1.1 (1):

“The area of longitudinal tension reinforcement should not be taken as less than “As, min”

$$A_{s,min} = 0,26 \frac{f_{ctm}}{f_{yk}} b_t d \quad (\text{but not less than } 0,0013b_t d)$$

Based on these equations, the minimum reinforcement comes out to be:

Minimum Reinforcement in (mm ²)		Using formula ($0.26 * f_{ctm} * b_t * d / f_{yk}$)	$0.0013b_t * d$	Minimum Area (Max of two)
	X+	465.2267	475.8	475.8
	X-	444.8889	455	455
	Y+	452.5156	462.8	462.8
	Y-	467.7689	478.4	478.4

Table 13: Minimum Reinforcement Requirement

$$A_{s, \max} = 0.04 * b * h = 0.04 * 1m * 0.4m = 16000 \text{ mm}^2$$

Euro code 2 - 9.3.1.1 (3):

The spacing of bars should not exceed S_{max} , slab:

$$S_{max} = \min (3h, 400mm)$$

$$3h = 3 * 400 \text{ mm} = 1200 \text{ mm};$$

Therefore;

$$S_{max} = \min (1200, 400) = 400 \text{ m}$$

Checking for maximum and minimum on the design reinforcement. Below tables shows area of steel provided. The area of steel in the table is in mm².

Asx+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	670	670	670	670	670	670	670	670	670	670	670	670	670
0.1	670	670	670	670	670	670	670	670	670	670	670	670	670
0.2	670	670	670	670	670	670	670	670	670	670	670	670	670
0.3	670	670	670	670	670	670	670	670	670	670	670	670	670
0.35	670	670	670	670	670	670	670	670	670	670	670	670	670
0.4	670	670	670	670	670					670	670	670	670
0.5	670	670	670	670	670					670	670	670	670
0.6	670	670	670	670	670					670	670	670	670
0.65	670	670	670	670	670					670	670	670	670
0.7	670	670	670	670	670	670	670	670	670	670	670	670	670
0.8	670	670	670	670	670	670	670	670	670	670	670	670	670
0.9	670	670	670	670	670	670	670	670	670	670	670	670	670
1.00	670	670	670	670	670	670	670	670	670	670	670	670	670

Table 14: Steel Area provided in bottom x-direction (mm²)

Asy+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	670	670	670	670	670	670	670	670	670	670	670	670	670
0.1	670	670	670	670	670	670	670	670	670	670	670	670	670
0.2	670	670	670	670	670	670	670	670	670	670	670	670	670
0.3	670	670	670	670	670	670	670	670	670	670	670	670	670
0.35	670	670	670	670	670	670	670	670	670	670	670	670	670
0.4	670	670	670	670	670					670	670	670	670
0.5	670	670	670	670	670					670	670	670	670
0.6	670	670	670	670	670					670	670	670	670
0.65	670	670	670	670	670					670	670	670	670
0.7	670	670	670	670	670	670	670	670	670	670	670	670	670
0.8	670	670	670	670	670	670	670	670	670	670	670	670	670
0.9	670	670	670	670	670	670	670	670	670	670	670	670	670
1.00	670	670	670	670	670	670	670	670	670	670	670	670	670

Table 15: Steel Area provided in bottom y-direction (mm²)

Asx-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	452	452	452	452	452	452	452	452	452	452	452	452	452
0.1	452	452	452	452	452	452	452	452	452	452	452	452	452
0.2	452	452	452	452	452	452	452	452	452	452	452	452	452
0.3	452	452	452	452	452	452	452	452	452	452	452	452	452
0.35	452	452	452	452	452					452	452	452	452
0.4	452	452	452	452	452					452	452	452	452
0.5	452	452	452	452	452					452	452	452	452
0.6	452	452	452	452	452					452	452	452	452
0.65	452	452	452	452	452	452	452	452	452	452	452	452	452
0.7	452	452	452	452	452	452	452	452	452	452	452	452	452
0.8	452	452	452	452	452	452	452	452	452	452	452	452	452
0.9	452	452	452	452	452	452	452	452	452	452	452	452	452
1.00	452	452	452	452	452	452	452	452	452	452	452	452	452

Table 16: Steel Area provided in top x-direction (mm^2)

Asy-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	452	452	452	452	452	452	452	452	452	452	452	452	452
0.1	452	452	452	452	452	452	452	452	452	452	452	452	452
0.2	452	452	452	565	565	565	452	452	452	452	452	452	452
0.3	452	452	452	565	565	754	452	452	452	452	565	565	452
0.35	452	452	452	565	565					452	565	565	452
0.4	452	452	452	565	565					452	565	565	452
0.5	452	452	452	565	565					452	565	565	452
0.6	452	452	452	565	565					452	565	565	452
0.65	452	452	452	565	452	452	452	452	452	452	565	565	452
0.7	452	452	452	452	452	452	452	452	452	452	452	452	452
0.8	452	452	452	452	452	452	452	452	452	452	452	452	452
0.9	452	452	452	452	452	452	565	565	565	565	565	565	452
1.00	452	452	452	452	452	452	565	565	565	565	565	565	452

Table 17: Steel Area provided in top y-direction (mm^2)

Based on the comparison of the values with the minimum and maximum area determined above, the design conforms to the code requirement.

Applying minimum spacing check:

The below tables represents spacing of design bars. The spacing mentioned below is in millimeters.

Asx+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	300	300	300	300	300	300	300	300	300	300	300	300	300
0.1	300	300	300	300	300	300	300	300	300	300	300	300	300
0.2	300	300	300	300	300	300	300	300	300	300	300	300	300
0.3	300	300	300	300	300	300	300	300	300	300	300	300	300
0.35	300	300	300	300	300	300	300	300	300	300	300	300	300
0.4	300	300	300	300	300					300	300	300	300
0.5	300	300	300	300	300					300	300	300	300
0.6	300	300	300	300	300					300	300	300	300
0.65	300	300	300	300	300					300	300	300	300
0.7	300	300	300	300	300	300	300	300	300	300	300	300	300
0.8	300	300	300	300	300	300	300	300	300	300	300	300	300
0.9	300	300	300	300	300	300	300	300	300	300	300	300	300
1.00	300	300	300	300	300	300	300	300	300	300	300	300	300

Table 18: Design Bar Spacing-Bottom x-direction

Asy+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	300	300	300	300	300	300	300	300	300	300	300	300	300
0.1	300	300	300	300	300	300	300	300	300	300	300	300	300
0.2	300	300	300	300	300	300	300	300	300	300	300	300	300
0.3	300	300	300	300	300	300	300	300	300	300	300	300	300
0.35	300	300	300	300	300	300	300	300	300	300	300	300	300
0.4	300	300	300	300	300					300	300	300	300
0.5	300	300	300	300	300					300	300	300	300
0.6	300	300	300	300	300					300	300	300	300
0.65	300	300	300	300	300					300	300	300	300
0.7	300	300	300	300	300	300	300	300	300	300	300	300	300
0.8	300	300	300	300	300	300	300	300	300	300	300	300	300
0.9	300	300	300	300	300	300	300	300	300	300	300	300	300
1.00	300	300	300	300	300	300	300	300	300	300	300	300	300

Table 19: Design Bar Spacing-Top y-direction

Asx-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	250	250	250	250	250	250	250	250	250	250	250	250	250
0.1	250	250	250	250	250	250	250	250	250	250	250	250	250
0.2	250	250	250	250	250	250	250	250	250	250	250	250	250
0.3	250	250	250	250	250	250	250	250	250	250	250	250	250
0.35	250	250	250	250	250					250	250	250	250
0.4	250	250	250	250	250					250	250	250	250
0.5	250	250	250	250	250					250	250	250	250
0.6	250	250	250	250	250					250	250	250	250
0.65	250	250	250	250	250	250	250	250	250	250	250	250	250
0.7	250	250	250	250	250	250	250	250	250	250	250	250	250
0.8	250	250	250	250	250	250	250	250	250	250	250	250	250
0.9	250	250	250	250	250	250	250	250	250	250	250	250	250
1.00	250	250	250	250	250	250	250	250	250	250	250	250	250

Table 20: Design Bar Spacing-Bottom x-direction

Asy-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	250	250	250	250	250	250	250	250	250	250	250	250	250
0.1	250	250	250	250	250	250	250	250	250	250	250	250	250
0.2	250	250	250	200	200	200	250	250	250	250	250	250	250
0.3	250	250	250	200	200	150	250	250	250	250	200	200	250
0.35	250	250	250	200	200					250	200	200	250
0.4	250	250	250	200	200					250	200	200	250
0.5	250	250	250	200	200					250	200	200	250
0.6	250	250	250	200	200					250	200	200	250
0.65	250	250	250	200	250	250	250	250	250	250	200	200	250
0.7	250	250	250	250	250	250	250	250	250	250	250	250	250
0.8	250	250	250	250	250	250	250	250	250	250	250	250	250
0.9	250	250	250	250	250	250	200	200	200	200	200	200	250
1.00	250	250	250	250	250	250	200	200	200	200	200	200	250

Table 21: Design Bar Spacing-Top y-direction

The comparison of the above table with code requirements shows that design steel conforms to the requirement.

Ultimate limit state (ULS) Check:

The resisting moments have been computed at ultimate limit state where following equations have been used. The equations have been derived by applying equilibrium to the concrete section under discussion at ultimate state when top concrete has reached ultimate strain (ϵ_{cu}).

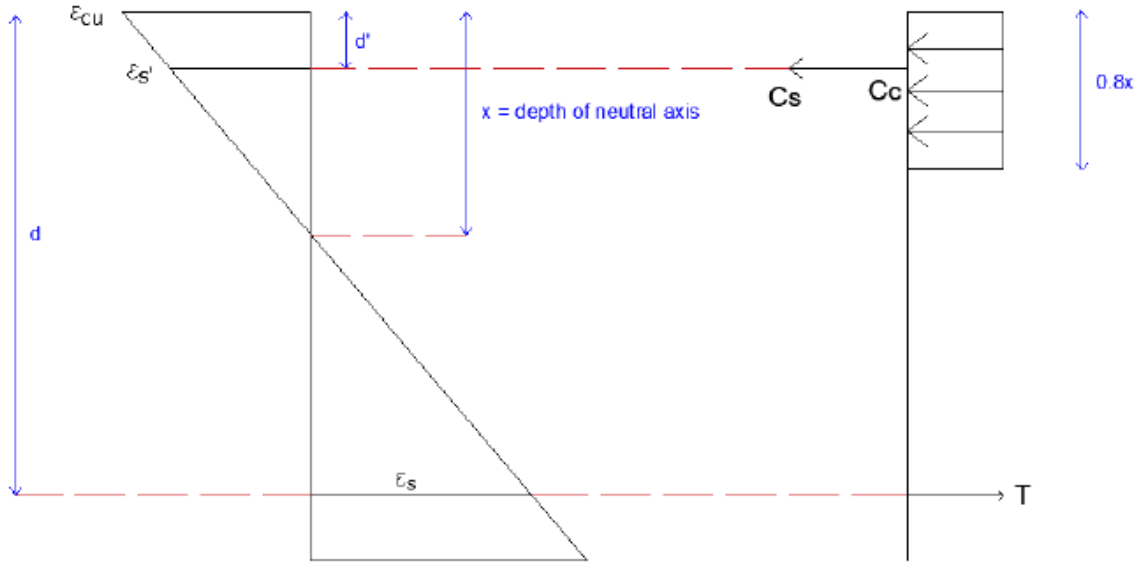


Figure 27: Beam Section for Ultimate Analysis

$$C_c + C_s + T = 0 \text{ ---- (i)}$$

$$M_r = T * Z_1 + C_s * Z_2 \text{ ---- (ii)}$$

Whereas,

C_c = Compression force carried by top concrete section

C_s = Compression force carried by top steel reinforcement

T = Tensile force carried by bottom reinforcement

$$Z_1 = d - 0.4x$$

$$Z_2 = 0.4x - d'$$

From eq (i), following equation is derived:

$$- 0.8 E + \lambda' * W_s' + \lambda * W_s = 0 \text{ ---- (iii)}$$

Whereas:

λ, λ' are ratio of respective steel strain with yield strain of steel. For this analysis λ is taken to be '1' as at ultimate state bottom steel is deemed to be yielded. However, the condition of λ' (top steel) is not known. Using following equation E is computed.

From which, using above strain geometry strain will be determined and its condition will be determined whether it is yielded or not.

$$0.8 E^2 - [(W_s + \alpha_o * W_s')] E + \alpha_o * \delta' * W_s' = 0$$

Where,

$$W_s = \frac{A_s * f_{yd}}{b * d * f_{cd}}, W_s' = \frac{A_{s'} * f_{yd}}{b * d * f_{cd}}, \alpha_o = x/d, \delta' = d'/d, E = x/d$$

Using above equation E is determined and the state of top reinforcement will be ascertained. Current equation assumes top reinforcement to be in elastic region. Once this assumption is ascertained, the results are used to determine resisting using eq(ii). If the assumption is not satisfied, the value of λ' is taken as unity and calculation is carried out accordingly. Based on the above discussion, following resisting moments have been determined:

Mx+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	90	90	90	90	90	90	90	90	90	90	90	90	90
0.1	90	90	90	90	90	90	90	90	90	90	90	90	90
0.2	90	90	90	90	90	90	90	90	90	90	90	90	90
0.3	90	90	90	90	90	90	90	90	90	90	90	90	90
0.35	90	90	90	90	90					90	90	90	90
0.4	90	90	90	90	90					90	90	90	90
0.5	90	90	90	90	90					90	90	90	90
0.6	90	90	90	90	90					90	90	90	90
0.65	90	90	90	90	90	-9	-9	-9	-9	90	90	90	90
0.7	90	90	90	90	90	90	90	90	90	90	90	90	90
0.8	90	90	90	90	90	90	90	90	90	90	90	90	90
0.9	90	90	90	90	90	90	90	90	90	90	90	90	90
1.00	90	90	90	90	90	90	90	90	90	90	90	90	90

Table 22: Resisting Moments-Bottom-x-direction

My+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	85	85	85	85	85	85	85	85	85	85	85	85	85
0.1	85	85	85	85	85	85	85	85	85	85	85	85	85
0.2	85	85	85	84	84	84	85	85	85	85	85	85	85
0.3	85	85	85	84	84	83	85	85	85	85	84	84	85
0.35	85	85	85	84	84					85	84	84	85
0.4	85	85	85	84	84					85	84	84	85
0.5	85	85	85	84	84					85	84	84	85
0.6	85	85	85	84	84					85	84	84	85
0.65	85	85	85	84	85	-5	-5	-5	-5	85	84	84	85
0.7	85	85	85	85	85	85	85	85	85	85	85	85	85
0.8	85	85	85	85	85	85	85	85	85	85	85	85	85
0.9	85	85	85	85	85	85	84	84	84	84	84	84	85
1.00	85	85	85	85	85	85	84	84	84	84	84	84	85

Table 23: Resisting Moments-Bottom-y-direction

Mx-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.1	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.2	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.3	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.35	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.4	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.5	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.6	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.65	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.7	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.8	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.9	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
1.00	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57

Table 24: Resisting Moments-Top-x-direction

My-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.1	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.2	-56	-56	-56	-70	-70	-70	-56	-56	-56	-56	-56	-56	-56
0.3	-56	-56	-56	-70	-70	-94	-56	-56	-56	-56	-70	-70	-56
0.35	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.4	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.5	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.6	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.65	-56	-56	-56	-70	-56	-56	-56	-56	-56	-56	-70	-70	-56
0.7	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.8	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.9	-56	-56	-56	-56	-56	-56	-70	-70	-70	-70	-70	-70	-56
1.00	-56	-56	-56	-56	-56	-56	-70	-70	-70	-70	-70	-70	-56

Table 25: Resisting Moments-Top-y-direction

The resisting moments obtained above are greater than the applied moments obtained through the Wood-Armer Method earlier.

The reinforcement layout and concerned detailing is given in the Annexure A given at the end of this report.

Following are the graphic representation of the results only for the sections in which there are the maximum absolute values of positive and negative acting moments.

Anchorage Length:

In order to ensure an opportune working of the reinforcements, we must compute the anchorage length for the steel bars. In fact, at the point of interruption, the tension within the bars will shatter inside the CLS.

Eurocode 2 – 8.4.1 (1):

“Reinforcing bars, wires or welded mesh fabrics shall be so anchored that the bond forces are safely transmitted to the concrete avoiding longitudinal cracking or spalling. Transverse reinforcement shall be provided if necessary.”

Defining l_{bd} as the design anchorage length, the minimum measure is:

In Tension:

$$l_{b,min} = \max \{0,3 \cdot l_{b,rqd}; 10 \varnothing; 100 \text{ mm}\}$$

In Compression:

$$l_{b,min} = \max \{0,6 \cdot l_{b,rqd}; 10 \varnothing; 100 \text{ mm}\}$$

Where l_{bd} is:

$$l_{bd} = \alpha_1 \cdot \alpha_2 \cdot \alpha_3 \cdot \alpha_4 \cdot \alpha_5 \cdot l_{b,rqd}$$

and,

$$l_{b,rqd} = \frac{\varnothing f_{yd}}{4 f_{bd}}$$

$$f_{bd} = 2,25 \cdot \eta_1 \cdot \eta_2 \cdot f_{ctd}$$

$$\eta_1 = \eta_2 = 1$$

Referring to the code following anchorage lengths are calculated for bent, loop and straight bars for bar $\varnothing 12$ and $\varnothing 16$.

The following table summarizes the results:

Zone	Diameter (mm)	Bar Type	C_d	$\alpha_{1,3,4,5}$	α_2	$l_{b,req}$ (mm)	l_{bd} (mm)	$l_{bd,min}$ (mm)	$l_{bd,selected}$ (mm)	$l_{bd,design}$ (mm)
Tension	12	Bent	25	1	1.14	522	415	157	415	600
Tension	12	Looped	25	1	1.14	522	415	157	415	600
Tension	12	Straight	25	1	0.84	522	306	157	306	450
Tension	16	Bent	25	1	1.22	696	592	209	592	900
Tension	16	Looped	25	1	1.22	696	592	209	592	900
Tension	16	Straight	25	1	0.92	696	446	209	446	650
Compression	12	Bent	25	1	1	522	365	313	365	550
Compression	12	Looped	25	1	1	522	365	313	365	550
Compression	12	Straight	25	1	1	522	365	313	365	550
Compression	16	Bent	25	1	1	696	487	417	487	700
Compression	16	Looped	25	1	1	696	487	417	487	700
Compression	16	Straight	25	1	1	696	487	417	487	700

Table 26: Anchorage Lengths

Request n.3

With reference to the reinforcement layout, assess the slab collapse load based on yield line theory and check the slab safety with respect to the design load considering different global and local mechanisms.

The below solution has been developed based on *Yield Line Theory* which determines the collapse load of the slab by the comparison of energy dissipated in the slab during the deformation (till formation of yield line) with external work done to dissipate the energy (the solution presented here does not consider energy losses not related to deformation). The balanced energy equation to be referred throughout the solution is given below.

$$\sum \int p_u(x, y) w(x, y) ds dA = \sum_{Y.L} \int m_{nu}(s) \theta_n(s) ds$$

The expression on the right represents the internal energy dissipation whereas the expression on the left represents the external work done.

External work:

$$\sum \int p_u(x, y) w(x, y) ds dA$$

The function $P_u(x, y)$ represents the collapse load. As the loading uniform for the both dimension (x and y) therefore, it can be kept constant which leaves $w(x, y)$ to be integrated over the area of the slab. Since the function w represents the deflection the integral will deliver volume of the deformed slab. Therefore, the integral represents the volume of the deformed slab.

External Work = Collapse Load * Volume of Deformed Slab = $P_{collapse} * V$.

Internal work:

$$\sum_{Y.L} \int m_{nu}(s) \theta_n(s) ds$$

The internal work can be computed by multiplying moments with corresponding rotation angle along the yield line.

Once the internal work is computed from above equation, the collapse load can be determined from following equation:

$$P_{collapse} = W_i / V$$

Since the Yield Line Method (Johansen) is an upper bound method, the load achieved with least possible value of above expression is selected as collapse load.

In the foregoing paragraphs, the computation of collapse load is carried out using above approach. There are two different mechanisms considered for the collapse; the Global Mechanism and Local Mechanism. For each of the two mechanisms, following procedure is adopted:

1. The yield line pattern is postulated
2. Volume of deformed slab is computed
3. Internal work has been computed
4. Determination of collapse load and comparison with design load

Since the computation of internal work requires resisting moments across the slab, therefore, the resisting moments obtained from Wood-Armer Method (Request n.2) have been reproduced below.

All of the moments given below are in KN.m/m.

Mx+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	90	90	90	90	90	90	90	90	90	90	90	90	90
0.1	90	90	90	90	90	90	90	90	90	90	90	90	90
0.2	90	90	90	90	90	90	90	90	90	90	90	90	90
0.3	90	90	90	90	90	90	90	90	90	90	90	90	90
0.35	90	90	90	90	90					90	90	90	90
0.4	90	90	90	90	90					90	90	90	90
0.5	90	90	90	90	90					90	90	90	90
0.6	90	90	90	90	90					90	90	90	90
0.65	90	90	90	90	90	-9	-9	-9	-9	90	90	90	90
0.7	90	90	90	90	90	90	90	90	90	90	90	90	90
0.8	90	90	90	90	90	90	90	90	90	90	90	90	90
0.9	90	90	90	90	90	90	90	90	90	90	90	90	90
1.00	90	90	90	90	90	90	90	90	90	90	90	90	90

My+	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	85	85	85	85	85	85	85	85	85	85	85	85	85
0.1	85	85	85	85	85	85	85	85	85	85	85	85	85
0.2	85	85	85	84	84	84	85	85	85	85	85	85	85
0.3	85	85	85	84	84	83	85	85	85	85	84	84	85
0.35	85	85	85	84	84					85	84	84	85
0.4	85	85	85	84	84					85	84	84	85
0.5	85	85	85	84	84					85	84	84	85
0.6	85	85	85	84	84					85	84	84	85
0.65	85	85	85	84	85	-5	-5	-5	-5	85	84	84	85
0.7	85	85	85	85	85	85	85	85	85	85	85	85	85
0.8	85	85	85	85	85	85	85	85	85	85	85	85	85
0.9	85	85	85	85	85	85	84	84	84	84	84	84	85
1.00	85	85	85	85	85	85	84	84	84	84	84	84	85

Mx-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.1	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.2	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.3	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.35	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.4	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.5	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.6	-57	-57	-57	-57	-57					-57	-57	-57	-57
0.65	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.7	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.8	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
0.9	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57
1.00	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57	-57

My-	x/a												
y/b	0	0.1	0.2	0.3	0.35	0.4	0.5	0.6	0.65	0.7	0.8	0.9	1
0	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.1	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.2	-56	-56	-56	-70	-70	-70	-56	-56	-56	-56	-56	-56	-56
0.3	-56	-56	-56	-70	-70	-94	-56	-56	-56	-56	-70	-70	-56
0.35	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.4	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.5	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.6	-56	-56	-56	-70	-70					-56	-70	-70	-56
0.65	-56	-56	-56	-70	-56	-56	-56	-56	-56	-56	-70	-70	-56
0.7	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.8	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56	-56
0.9	-56	-56	-56	-56	-56	-56	-70	-70	-70	-70	-70	-70	-56
1.00	-56	-56	-56	-56	-56	-56	-70	-70	-70	-70	-70	-70	-56

Table 27: Resisting Moments for Yield Analysis

Since the determination of internal work depends on resisting moments, therefore before proceeding ahead with the solution, the slab has been divided into sections wherever bending moments vary to assist in the development of the solution.

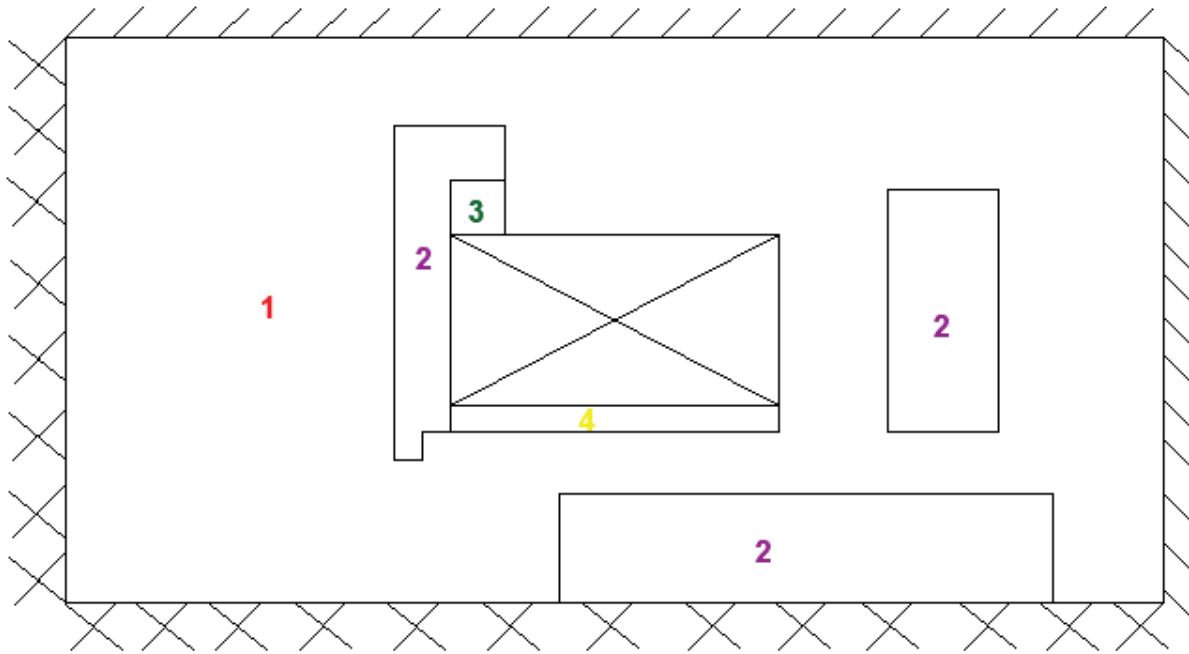


Figure 28: Slab regions for Yield Line Analysis

The slab is divided into three regions due to varying M_y^+ and M_y^- moments across the slab. Since moments vary only in x-direction, therefore only requisite dimensions are shown (all dimensions are in millimeters).

Based on the above division, the slab moments values for each region are given below:

	Region1	Region2	Region 3	Region 4
M_x^+	90	90	90	-9
M_y^+	85	84	83	-5
M_x^-	-57	-57	-57	-57
M_y^-	-56	-70	-94	-56

Table 28: Summarized Resisting Moments for each region

All of the above values are in KN.m/m . For the coming paragraphs the positive and negative moments along x and y directions for a particular section will be notified by region number at the end of notation (i.e. $M_{x,1}^+$; Positive moment in x-direction in section 1).

Global Mechanism:

In order to proceed with the computation of collapse load, the below yield line pattern is postulated. The pattern represents negative and positive yield lines. The negative yield line will appear at the fixed edges where negative moments are higher whereas positive yield line travel through the slab as the positive bending moments are higher inside of the supports. The depiction of the yield line pattern is given below. In order to better depict the yield lines, the positive and negative yield lines are shown separately:

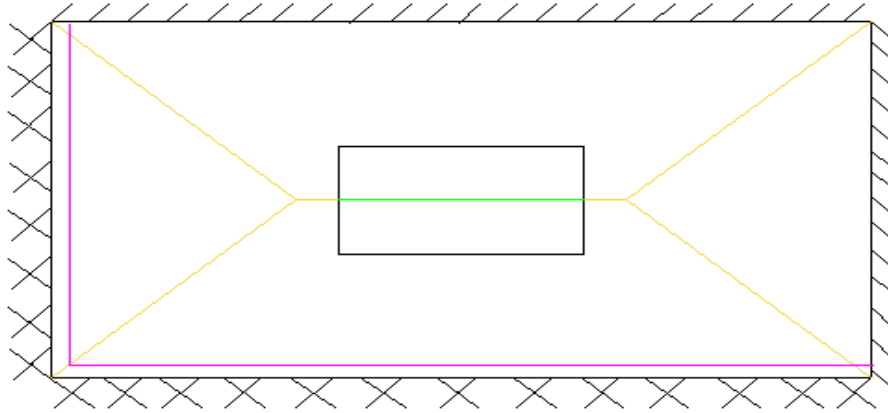


Figure 29: Positive and Negative yield line

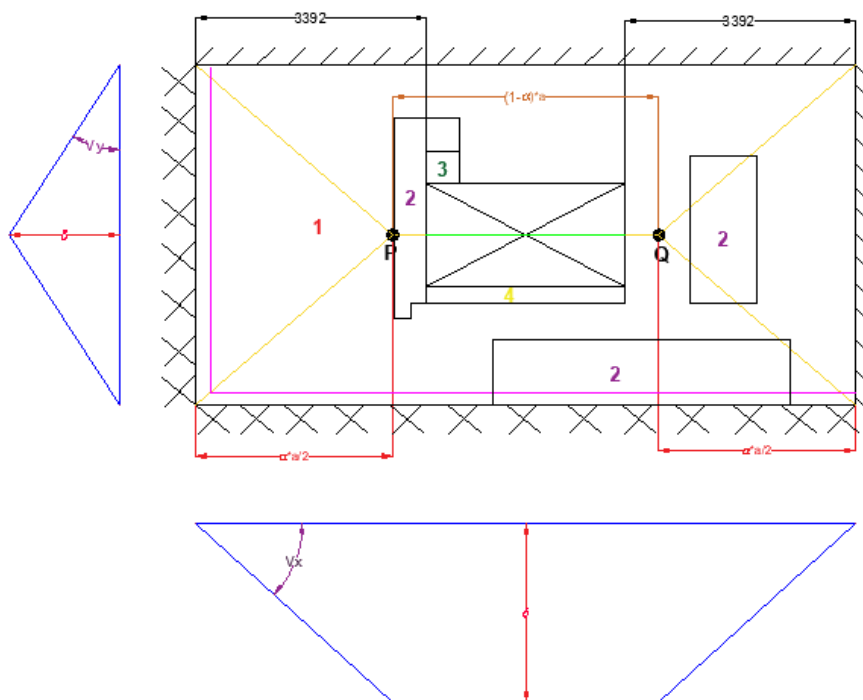


Figure 30: Yielded Slab Notations



In addition to the yield line, certain parameters are also shown in the above picture. The major one is $\alpha \cdot a/2$ which shows the horizontal extent of the point P and Q which are the intersection points of the respective positive yield lines. The positions of the P and Q is unknown and depends on the factor α . During the solution, the ratio of Internal Work (W_i) and Volume (V) (i.e., the collapse load) will be obtained in the terms of the ' α ' and the value corresponding to the least collapse load will be selected to determine the collapse load. This will be done by equating the derivate of W_i/V to zero and determining the corresponding value of ' α '. The value of α can range from 0 to 1. This is due to fact that the Yield Line Method (Johansen) is an upper bound method and requires determination of least value of operating parameter to approximate the value of minimum collapse load.

$$d(W_i/V)/d\alpha = 0$$

In addition to this, the horizontal and vertical section of deformed slab is show (green sections). The maximum deflection of the slab is denoted by letter ' δ '. The value of curvature V_x and V_y are given below:

$$V_x = \delta/(\alpha \cdot a/2)$$

$$V_y = \delta/(a/2)$$

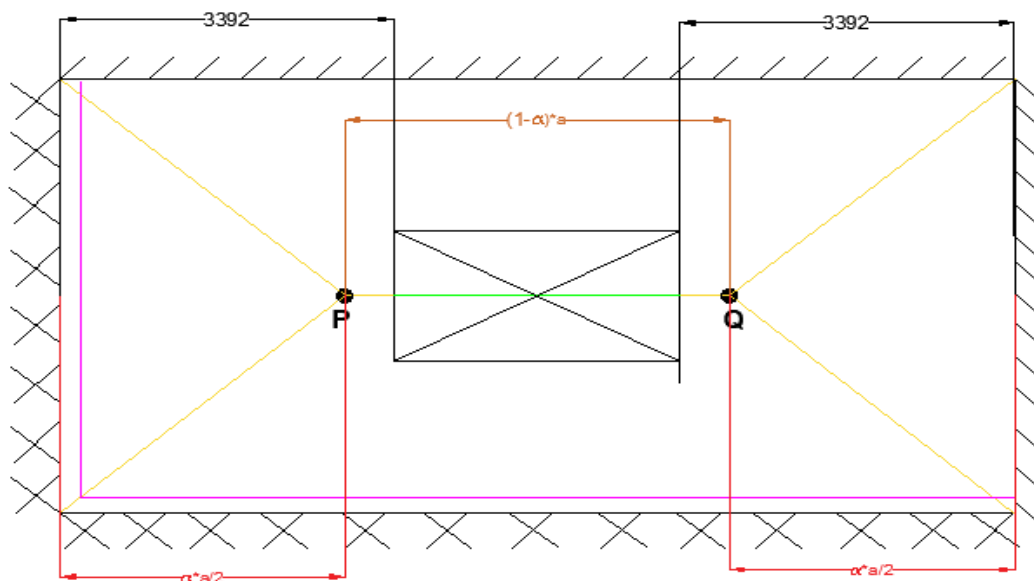
To proceed with solution the expression of V and W_i are to be determined. Both are carried out separately.

Volume (V):

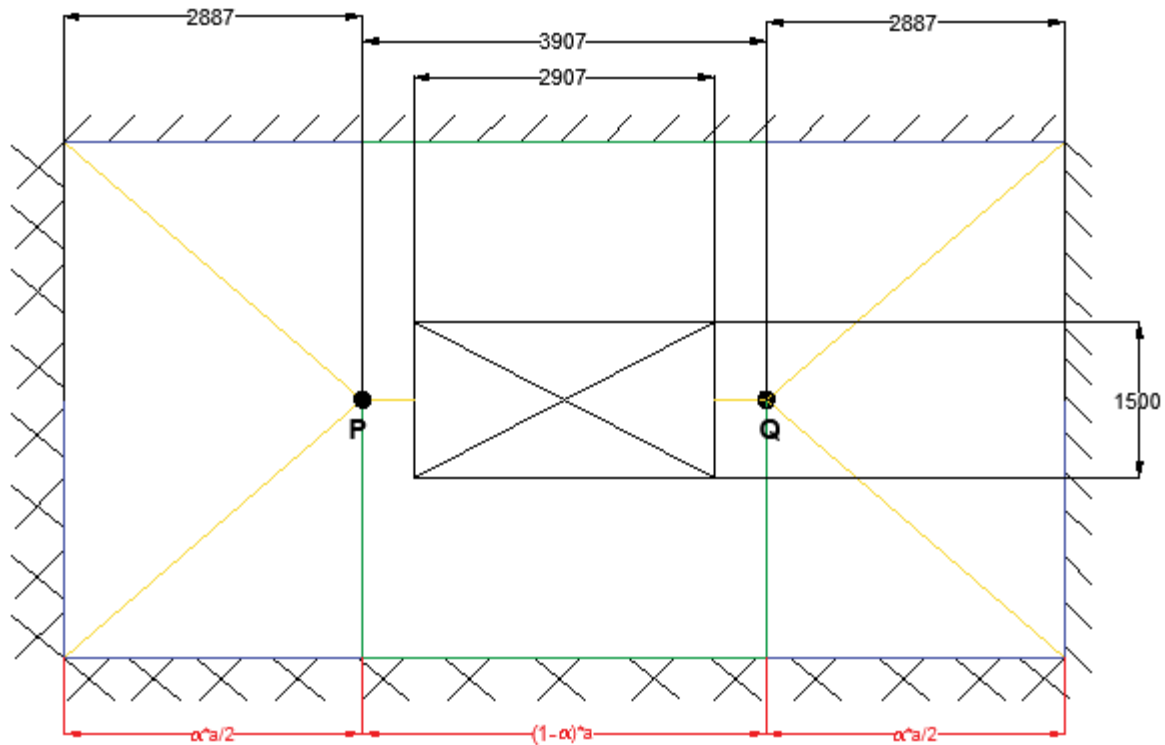
Since the location of the Point P and Q are unknown and given that slab geometry is non-symmetric with respect to vertical axis, the volume equation is derived for following two cases:

Case 1 ($\alpha \cdot a/2 < 3.392$ m):

The case represents the scenario when Point P & Q lies within right portion of slab of width 3392 mm (shown in figure below).



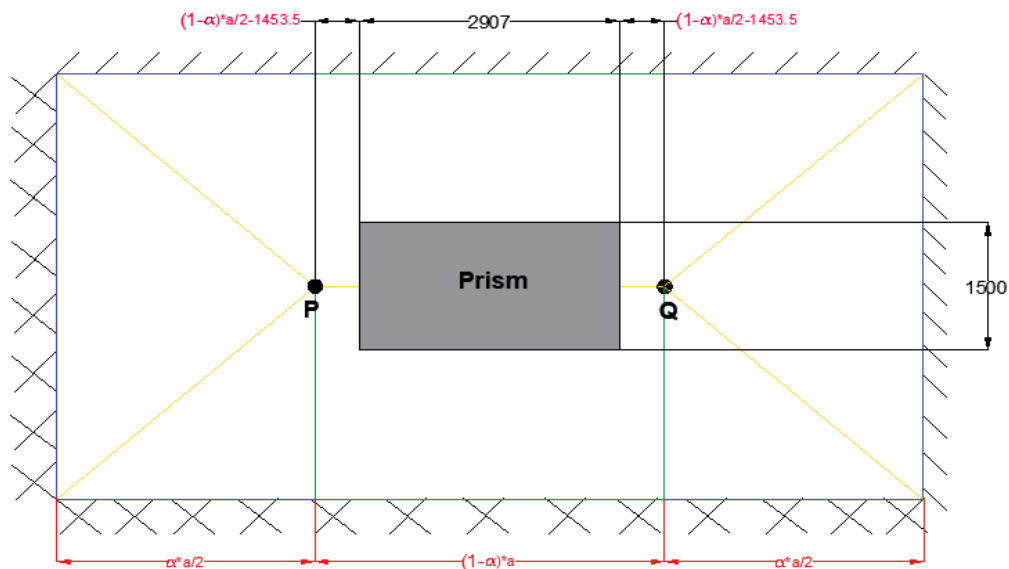
In order to determine the volume the slab can be divided into two pyramids (left of Point P and right of Point Q) and one prism (area between Point P and Q) as shown below. The blue rectangles in the below figure represent pyramids whereas green presents prism.



Next, the imaginary slab portions (shown in Grey color in figure below) will be subtracted from the sum of aforementioned three figures. It shall be noticed that the imaginary surfaces do not exist and have counted in the above mentioned sum. The imaginary surfaces consists of one prisms.

Therefore, the final equation is:

$$V = (2 * \text{Area of Pyramid} + \text{Area of Prism}) - (\text{Area of Imaginary Prism})$$

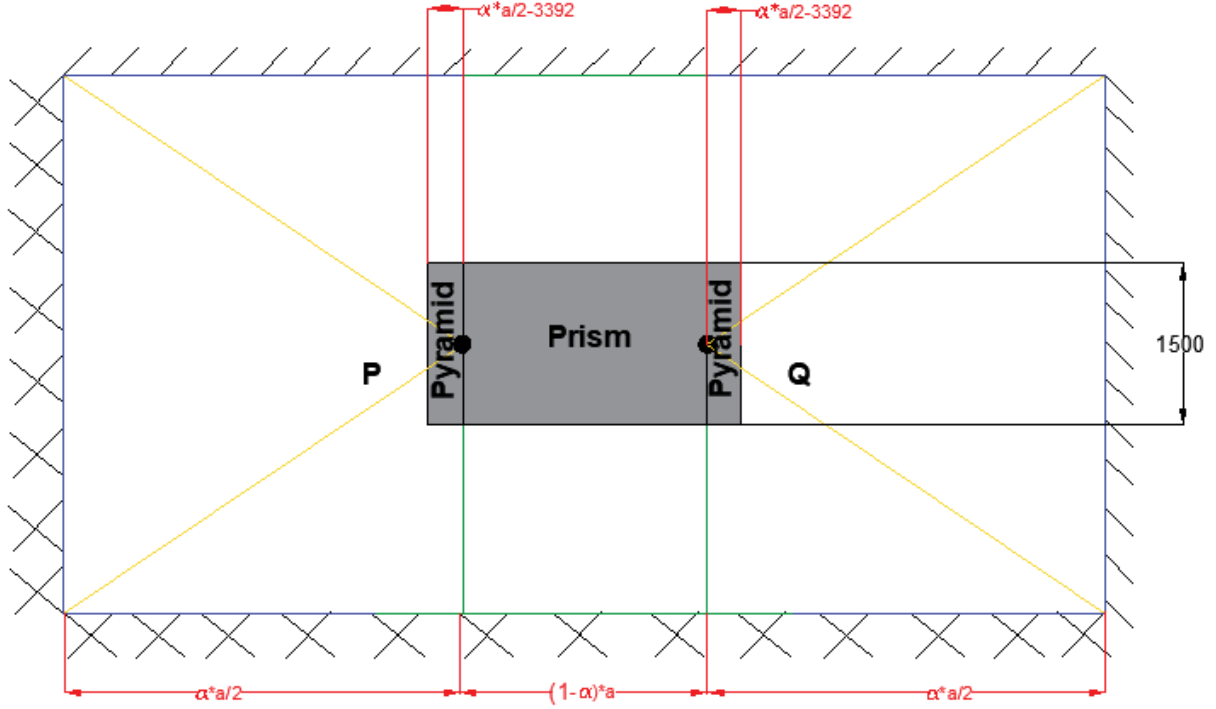


$$V = (2*(1/3*\alpha*a/2*5*\delta) + (1/2*(1-\alpha)*a*5*\delta)) - (1/2*1.5*2.907*(3.7*\delta/a))$$

$$V = \frac{5*\alpha*a*\delta}{3} + \frac{5*(1-\alpha)*a*\delta}{2} - \frac{8.1*(1-\alpha)*\delta}{a}$$

Case 2 ($\alpha*a/2 > 3.392$ m):

The same calculation will be repeated for this case except for the imaginary where one prism and two Pyramid will be left. This can be more clearly seen from the following figure.



Using similar approach, the volume comes out to be:

$V = (2*\text{Area of Pyramid} + \text{Area of Prism}) - (\text{Area of Imaginary Prism}) - (2*\text{Area of Imaginary Prism})$

$$V = (2*(1/3*\alpha*a/2*5*\delta) + (1/2*(1-\alpha)*a*5*\delta)) - (1/2*1.5*(1-\alpha)*a*(3.7*\delta/a)) -$$

$$2*(1/3*1.5*(\alpha*a/2-3.392)*(3.7*\delta/a))$$

$$V = \frac{5*\alpha*a*\delta}{3} + \frac{5*(1-\alpha)*a*\delta}{2} - \frac{5.55*(1-\alpha)*\delta}{2} - \frac{12.55*\delta}{a} + \frac{3.7*\delta*\alpha}{2}$$

Therefore, we have two equations for volume (V) and will be applied for corresponding cases (i.e., for $\alpha*a/2 > 3.392$ m and $\alpha*a/2 < 3.392$ m).

Having calculated volume of the deformed slab, now the internal work will be computed to reach the final expression of $P(\alpha)$.

Internal work:

The equation for calculation of internal work required resisting moment along yield line, curvature and length of corresponding yield line.

$$W_i = M * \theta * L$$

Considering the postulated yield pattern, the calculation of W_i for each yield line will be carried out. Since $\alpha*a/2$ is unknown at the present time and also the moments vary in this direction as well (Section 1, 2, 3 and 4 where M_y^+ and M_y^- are different). Therefore, the calculation of W_i will be carried out in multiple intervals so that these variations can be accommodated in the calculation. Also the geometry of the slab will require intervals to be define as volume equation differs whether $\alpha*a/2$ is greater or less than 3.392 m.

Therefore, keeping above in consideration, the slab is divided into following intervals. The calculation related to negative yield line will remain same for each interval as it does not depend on factor ' α '.

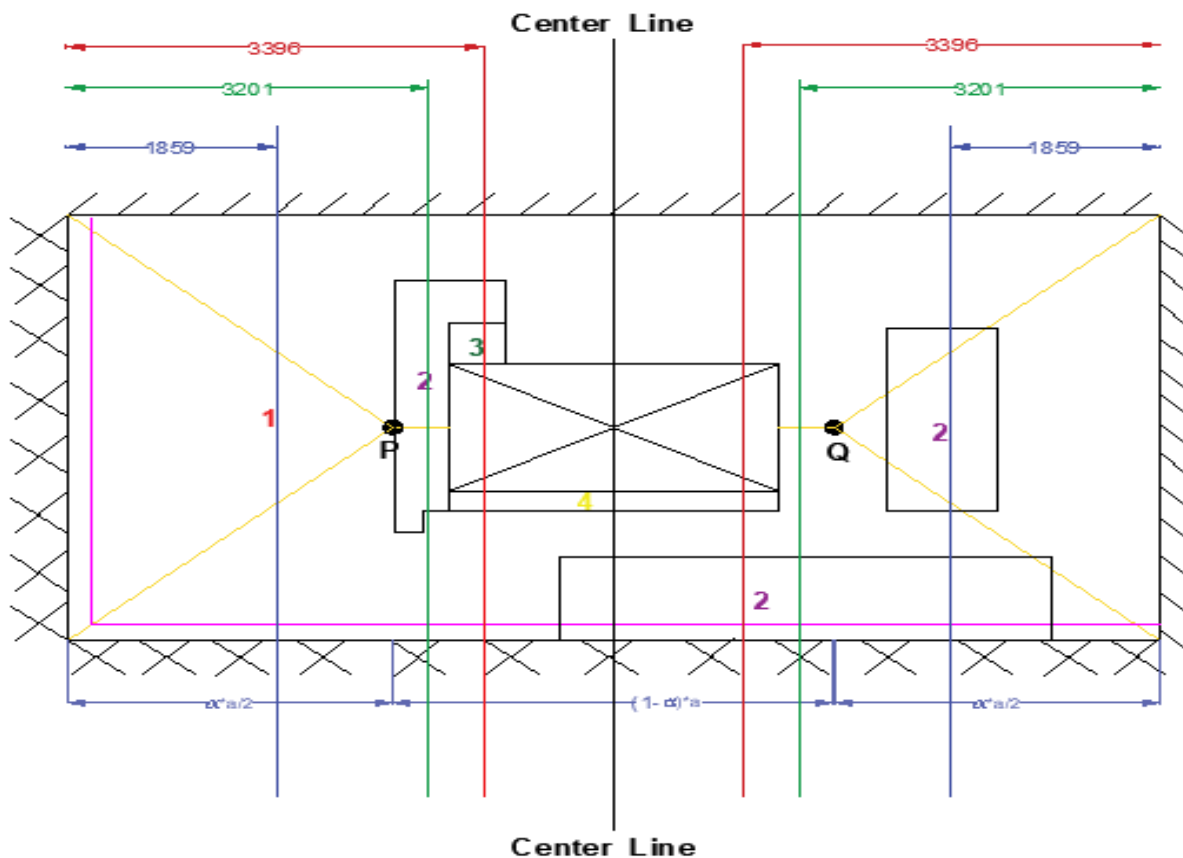


Figure 31: Slab Interval for analysis-Global Collapse Mechanism

Based on the geometry of the slab and moment distribution the slab is divided into three intervals shown above. The range of the intervals is given below:

- Interval 1: $0 < \alpha \cdot a/2 < 1.859$ (length of interval ($L1 = 1.859$))
- Interval 2: $1.859 < \alpha \cdot a/2 < 3.201$ (length of interval ($L2 = 1.342$))
- Interval 3: $3.201 < \alpha \cdot a/2 < 3.396$ (length of interval ($L3 = 0.195$))

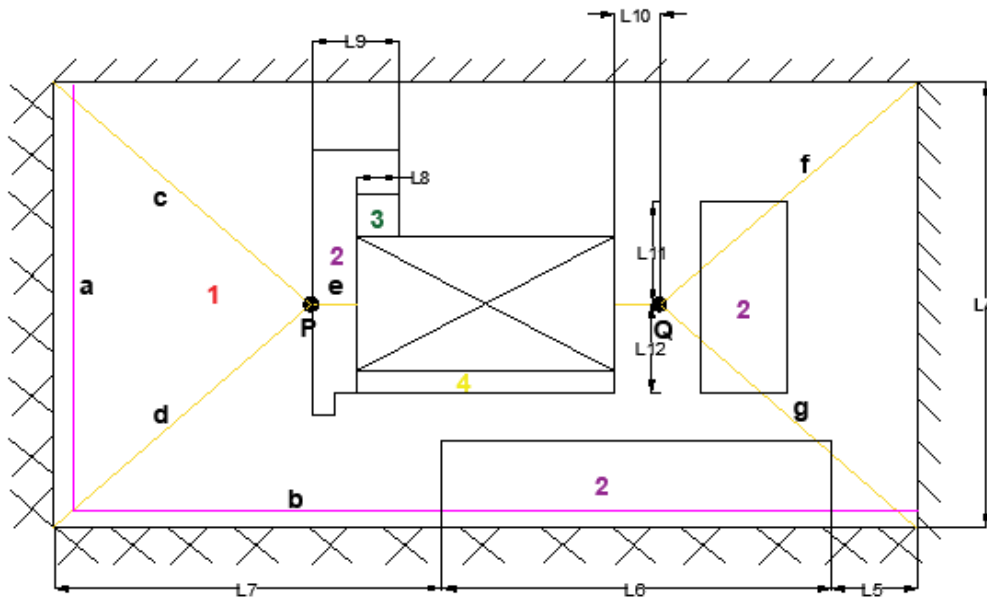
The region beyond Interval 3 and before the central line is very small and was not investigated as convergence was reached at the Interval 3.

Based on these intervals, the calculation has been carried out. The case 1 of volume (V) is used for Interval 1 whereas case 2 of volume (V) is used for Interval 2 and 3 given the geometry of intervals.

The calculation here is only shown for Interval 1 whereas for Interval 2 and 3 the main calculation is not scribed here and only expression $d(W_i/V)/d\alpha$ and subsequent calculation is given:

Interval 1:

For the sake of reference, the yield lines have been labeled as shown in below picture:



$W_i = M \cdot v \cdot L$; to be computed for Yield Line a, b, c, d, e, f and g

Yield Line – (a):

$$= (M_{y,1}^- \cdot v_y \cdot L4)$$

Yield Line – (b):

$$= (M_{y,1}^- \cdot v_y \cdot L5) + (M_{y,2}^- \cdot v_y \cdot L6) + (M_{y,1}^- \cdot v_y \cdot L7)$$

Yield Line – (c):

$$= (M_{x,1}^+ * v_x * L4/2) + (M_{y,1}^+ * v_x * \alpha * a/2)$$

Yield Line – (d):

$$= (M_{x,1}^+ * v_x * L4/2) + (M_{y,1}^+ * v_x * \alpha * a/2)$$

Yield Line – (e):

$$= \{M_{x,2}^+ * v_x * (L9-L8)\} + (M_{x,1}^+ * v_x * L10)$$

Yield Line – (f):

$$= \{M_{x,1}^+ * v_x * (L4/2-L11)\} + (M_{x,2}^+ * v_x * L11)$$

Yield Line – (g):

$$= \{M_{x,1}^+ * v_x * (L4/2-L12)\} + (M_{x,2}^+ * v_x * L12)$$

$$W_{internal} = \sum \text{Yield line}$$

Putting following values:

$$M_{y,1}^- = -56 \frac{KN.m}{m}$$

$$M_{y,2}^- = -70 \frac{KN.m}{m}$$

$$M_{x,1}^+ = 90 \frac{KN.m}{m}$$

$$M_{y,1}^+ = 85 \frac{KN.m}{m}$$

$$M_{x,2}^+ = 90 \frac{KN.m}{m}$$

$$L1=1.859m$$

$$L2=1.342m$$

$$L3=0.195m$$

$$L4=5m$$

$$L5=0.969m$$

$$L6=3.876m$$

$$L7=4.845m$$

$$L8=0.485m$$

$$L9=0.969m$$

$$L10=0.969m$$

$$L11=1m$$

$$L12=0.75m$$

$$a=9.69m$$

$$v_x = 2\delta/\alpha \cdot a = 2\delta/\alpha \cdot 7.4$$

$$v_y = 2\delta/a = 2\delta/7.4$$

Putting all these into above equation the result is:

$$W_i = \delta (243.4 + 180.96/\alpha)$$

Using volume equation for Case 1, volume comes out to be:

$$V = \delta (-7.9875\alpha + 24.98425)$$

$$W_i/V = (243.4\alpha + 180.96)/(-7.9875\alpha^2 + 24.98425\alpha)$$

Taking derivative of above expression and putting the output equal to 0, the α comes out to be 0.95 which for this case should be less than 0.3 (i.e., $1.11/(7.4/2)$).

Therefore, proceeding ahead with the next interval with same method, the expression W_i/V comes out to be:

For Interval 2:

$$W_i/V = (243.4 + 180.96/\alpha)/(-7.9875\alpha^2 + 24.98425\alpha)$$

Equating the derivative of above expression to 0, the value of α comes out to be 1.65 which is not converging as α should be less than or equal to 1.

Therefore proceeding ahead with same method:

For Interval 3:

$$W_i/V = (243.4 + 180.96/\alpha)9.1266\alpha^2 + 25.7828\alpha$$

Equating the derivative of above expression to 0, the value of α comes out to be 0.84 which is converging as value of α in this interval shall be between 0.7 and 0.9

Based on this, the collapse load comes out to be:

$$\text{Minimum Collapse Load} = 37.2 \text{ kN/m}^2$$

The ratio of collapse load with design load (21.825 kN/m^2) is:

$$= 37.2 / 21.825$$

$$= 1.7 > 1.5$$

The minimum collapse load passes the check.

Local Mechanisms:

Based on the geometry, support conditions and applied moments, the following local failure mechanisms are studied:

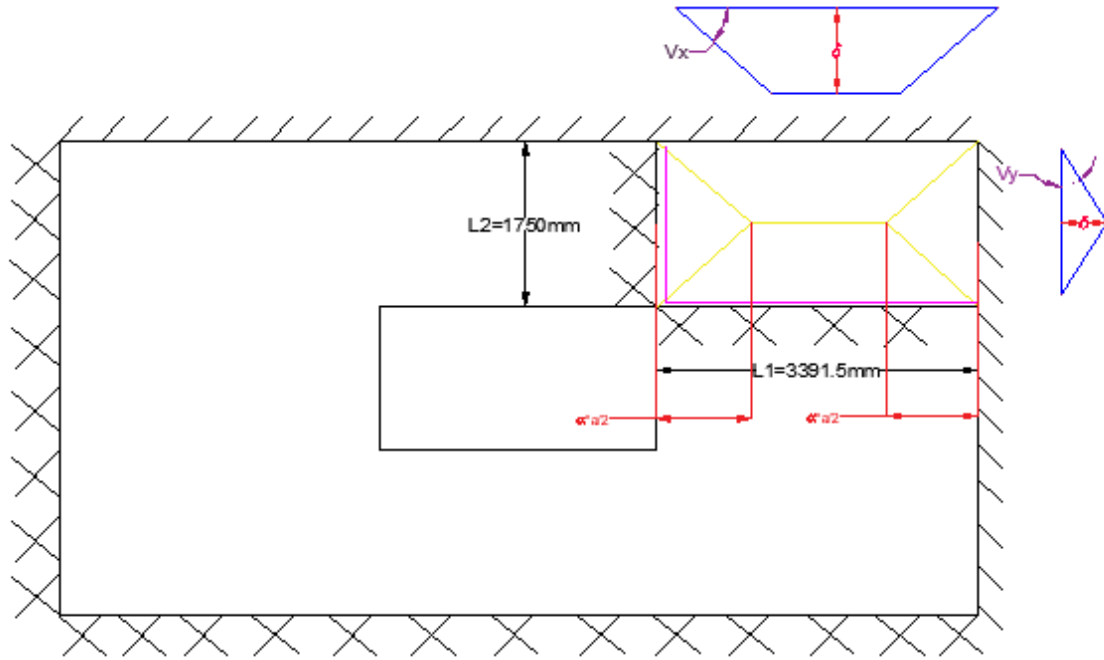
Local Mechanism 1:

Figure 32: Yield Line Analysis-Local Mechanism

Volume (V):

$$= 2 * (1/3 * \alpha * \delta * L_2 * L_1 / 2) + (1/2 * (1 - \alpha) * \delta * L_2 * L_1)$$

Using following:

$$L_1 = 3.3915 \text{ m}; L_2 = 1.75 \text{ m}$$

$$V = \delta (-2.966 * \alpha + 8.8985)$$

Internal Work:

$$v_x = 2\delta / \alpha * L_1$$

$$v_y = 2\delta / L_2$$

$$W_i = 2 * (M_{y,3}^- * v_y * L_1) + (M_{x,3}^- * v_x * L_2) + 2 * (M_{x,3}^+ * v_x * L_2) + 2 * (M_{y,3}^+ * v_y * \alpha L_1 / 2 * 2) + (M_{y,3}^+ * v_y * 2 * (1 - \alpha) * L_1)$$

Using lengths given above and moment values given in the global mechanism section the expression W_i comes out to be:

$$W_i = \delta (243.4 + 180.96 / \alpha)$$

$$W_i / V = \delta (243.4 + 180.96 / \alpha) / (-2.966 * \alpha^2 + 8.8985 \alpha)$$

Taking derivative of above expression and equating to gives:

$$\alpha = 0.817$$

This gives the value of minimum collapse load to be:

$$p = 23.066 > p_{global} > p_{applied}$$

Local Mechanism 2:

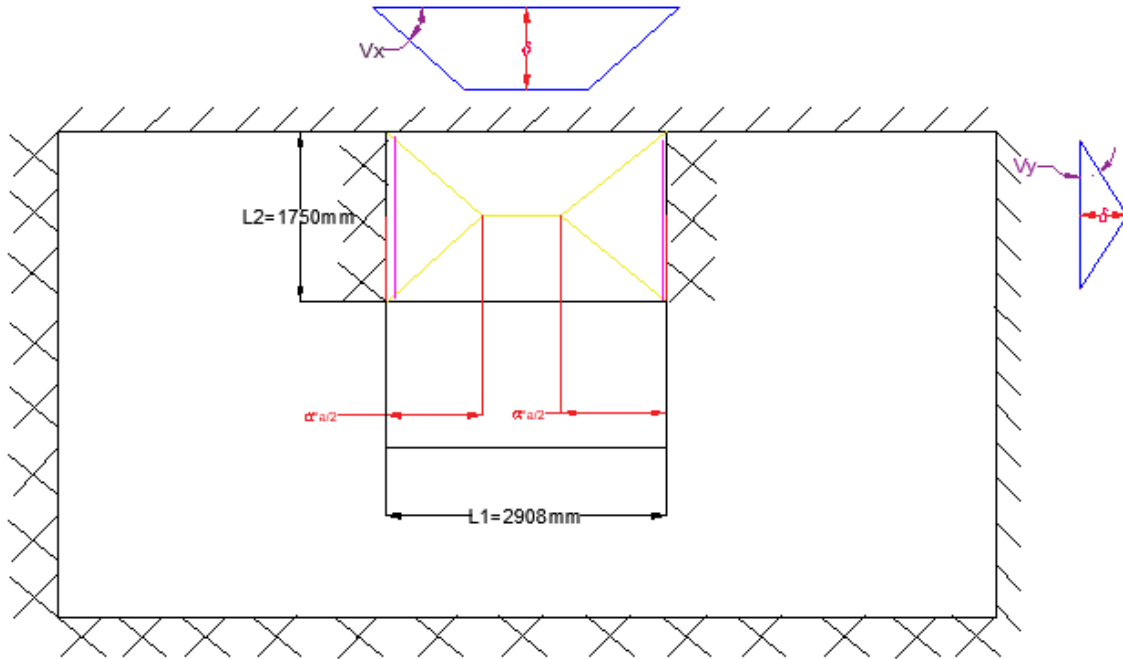


Figure 33: Yield Line Analysis-Local Mechanism

At the unsupported edge, the yield line directly snaps at slab edge as there is no slab support providing bending resistance.

Volume (V):

$$= (1/3 * \alpha * \delta * L_2 * L_1) + (1/2 * (1 - \alpha) * \delta * L_2 * L_1)$$

Using following:

$$L_1 = 2.908 \text{ m}; L_2 = 1.75 \text{ m}$$

$$V = \delta (-1.48325 * \alpha + 4.44925)$$

Internal Work:

$$v_x = 2\delta / \alpha * L_1$$

$$v_y = 2\delta / L_2$$

$$W_i = (M_{y,3}^- * v_y * L_1) + (M_{x,3}^+ * v_x * L_2) + 2 * (M_{y,3}^+ * v_y * L_2) + (M_{y,3}^+ * v_y * (1 - \alpha) * L_1)$$

Using lengths given above and moment values given in the global mechanism section the expression W_i comes out to be:

$$W_i = \delta (503.07 + 23.199/\alpha + 302.4\alpha)$$

$$W_i/V = (503.07\alpha + 23.199 + 302.4\alpha^2) / (-1.48325*\alpha^2 + 4.44925\alpha)$$

Taking derivative of above expression and equating to gives:

$$\alpha = 0.2$$

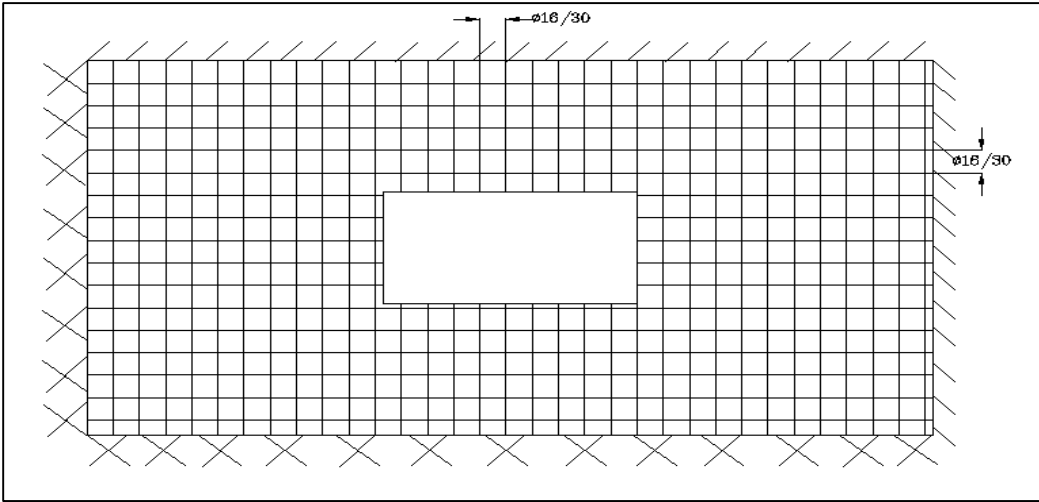
This gives the value of minimum collapse load to be:

$$p = 56.36172 > p_{global} > p_{applied}$$

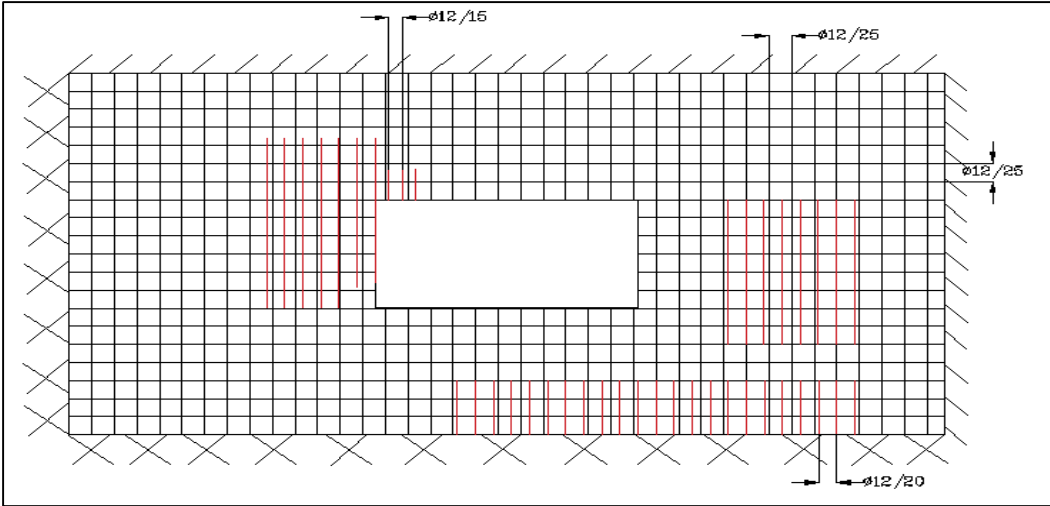
In conclusion, from the results obtained in terms of local mechanisms, it can be stated that every mechanism passes the safety check of the slab, and so with respect to the investigated cases here the plate would present a global collapse rather than a local one. It is however clear that, only a small number of configurations have been considered and assessment of further mechanisms may provide varying trend.

ANNEXURES

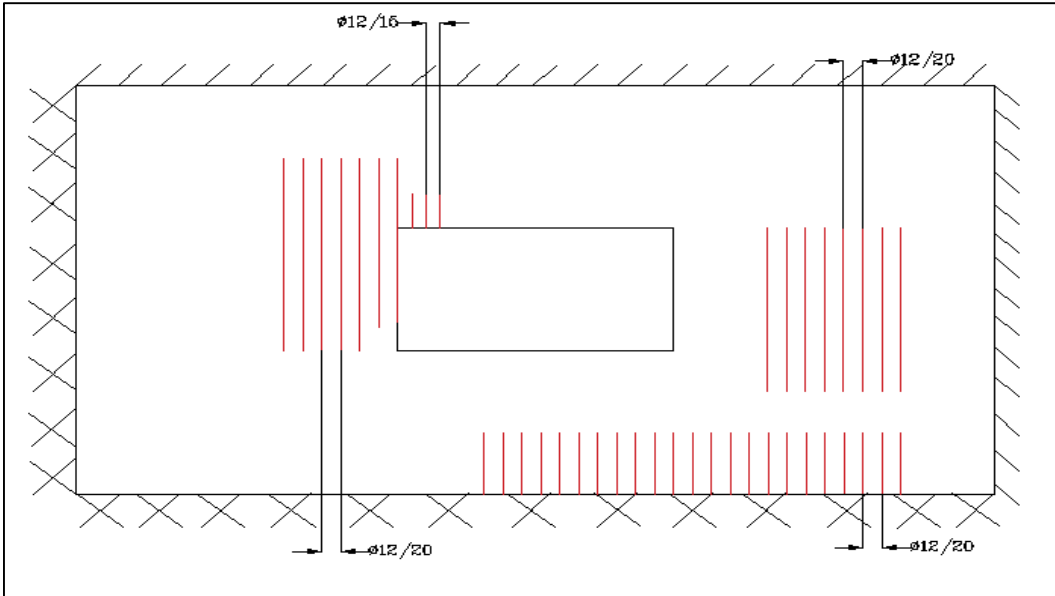
Bottom Steel Reinforcement Layout

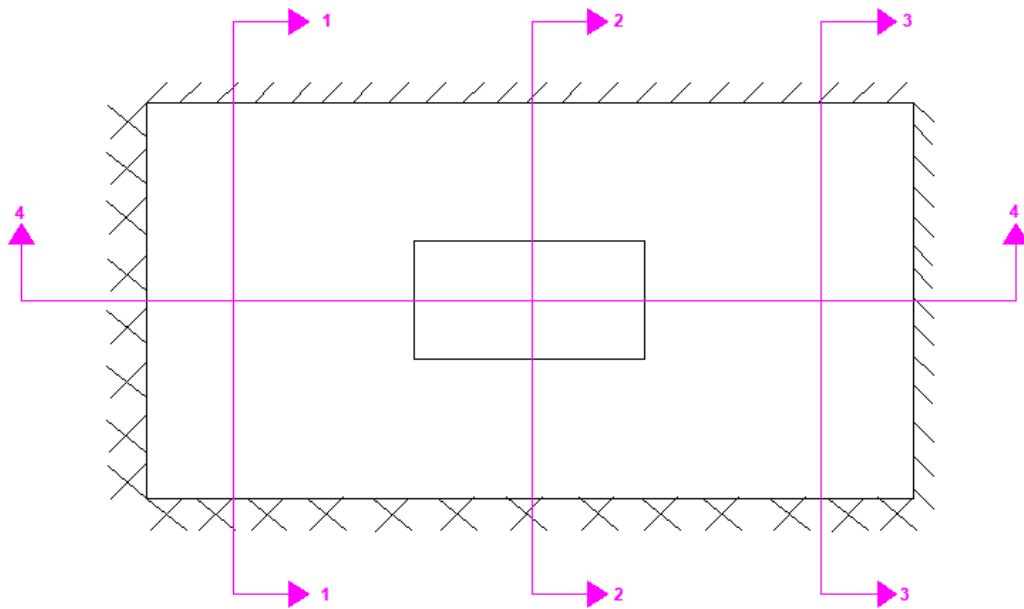


Top Steel Reinforcement Layout

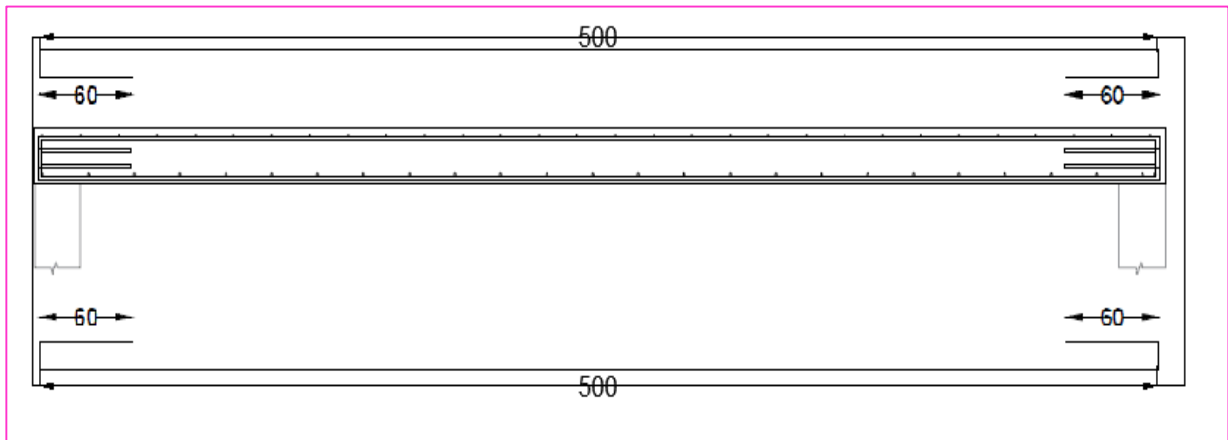


Top Steel Reinforcement Layout Refinement

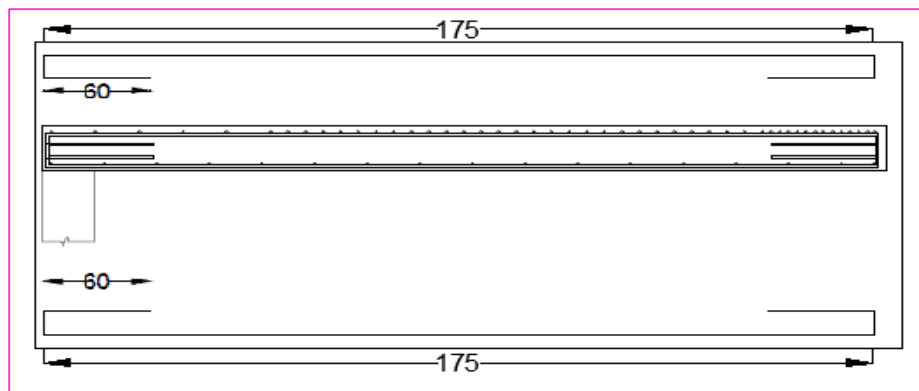




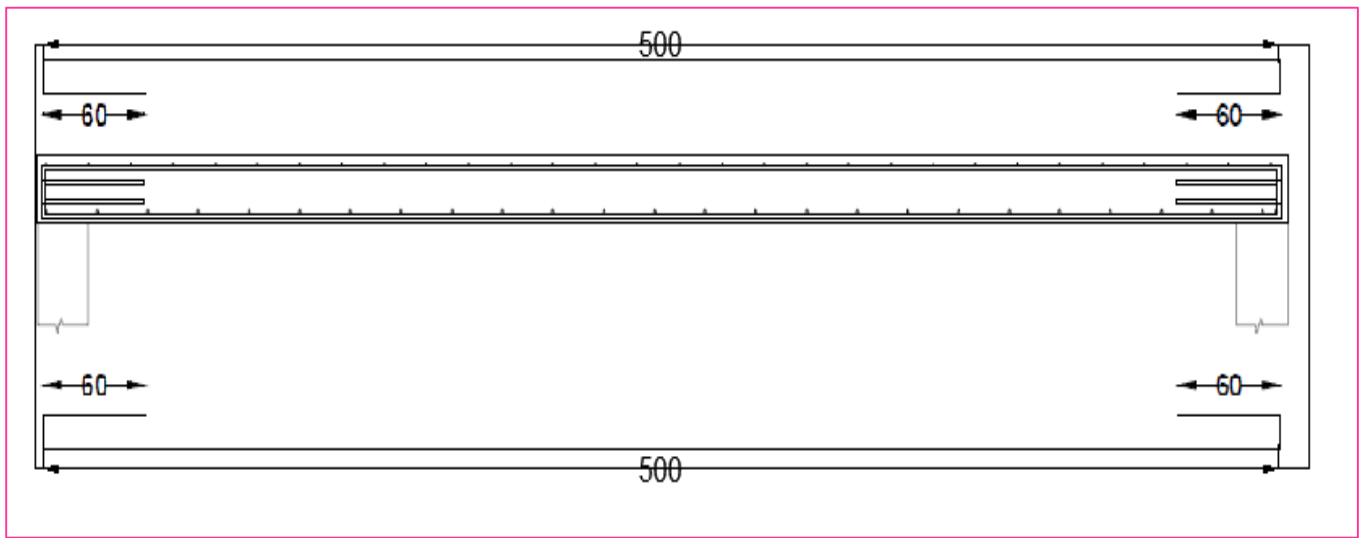
Section 1-1



Section 2-2



Section 3-3



Section 4-4

